



BITS Pilani
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Automotive Communication and Networking Lab Experiment 3 – Headlamp glowing using MBD

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Exercise-1

Exercise - 1

Glowing of LED by transmitting a CAN message in smart wheels V1 Pro with TC1001.

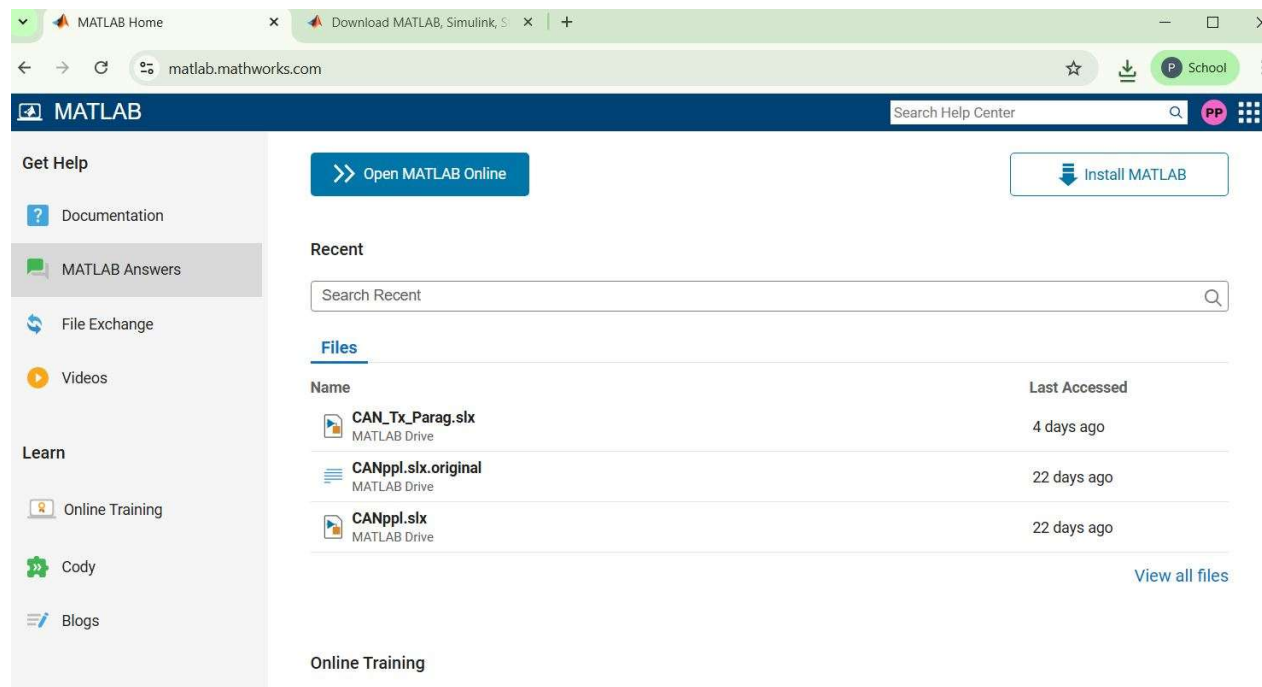
Aim: To glow LED by transmitting different CAN messages in smart wheels V1 Pro with TC1001 .

Hardware Requirements: Smart wheels V1 Pro with TC1001.

Software Requirements: MATLAB Simulink with NXP support package S32K1xx

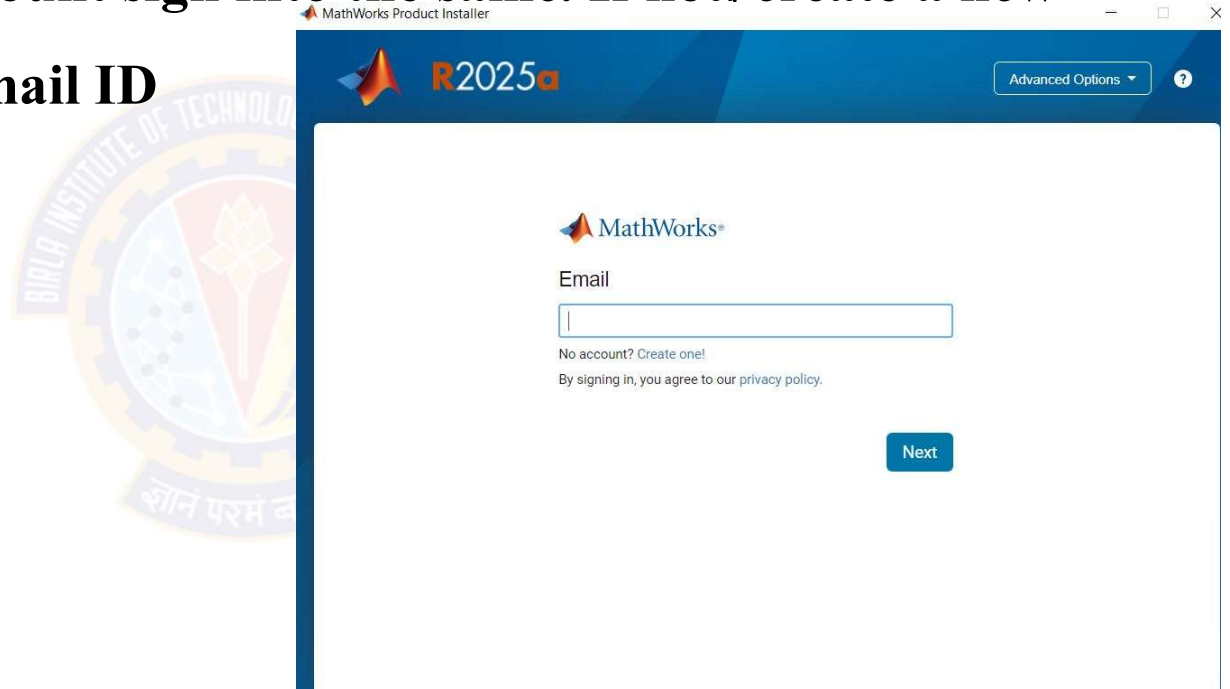
Installing MATLAB

- ❑ Go to <https://matlab.mathworks.com/> and click Install MATLAB



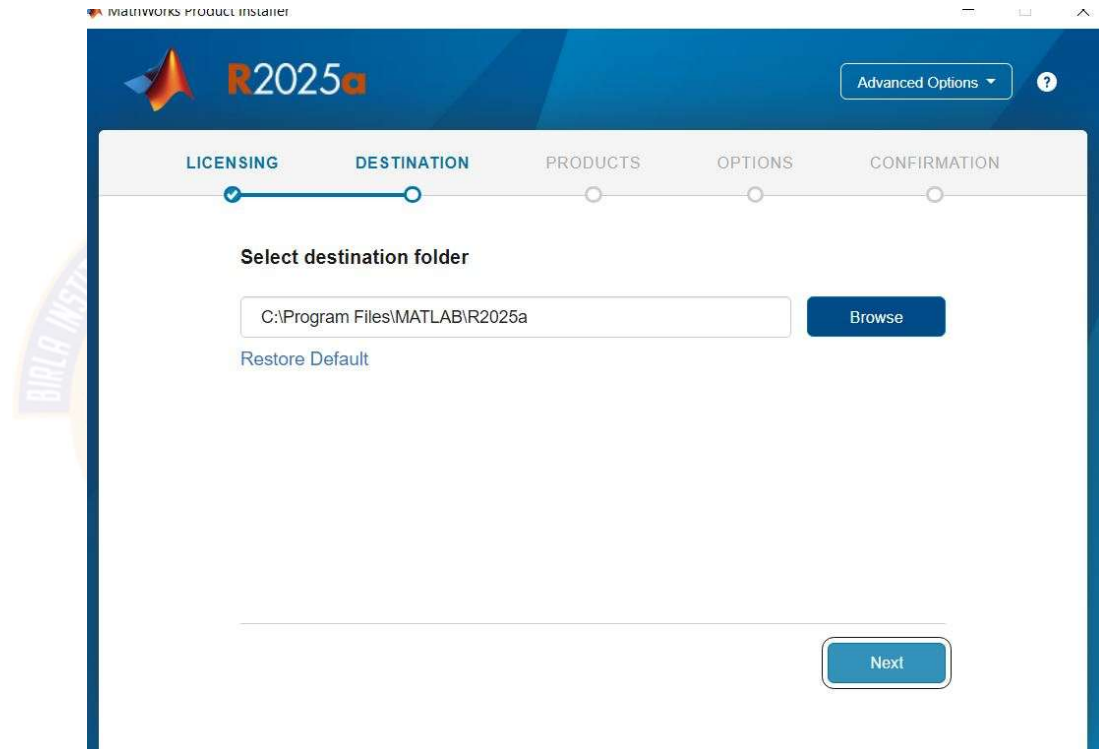
Installing MATLAB

- ❑ If you have a MathWork account sign into the same. If not, create a new account using your BITS Email ID



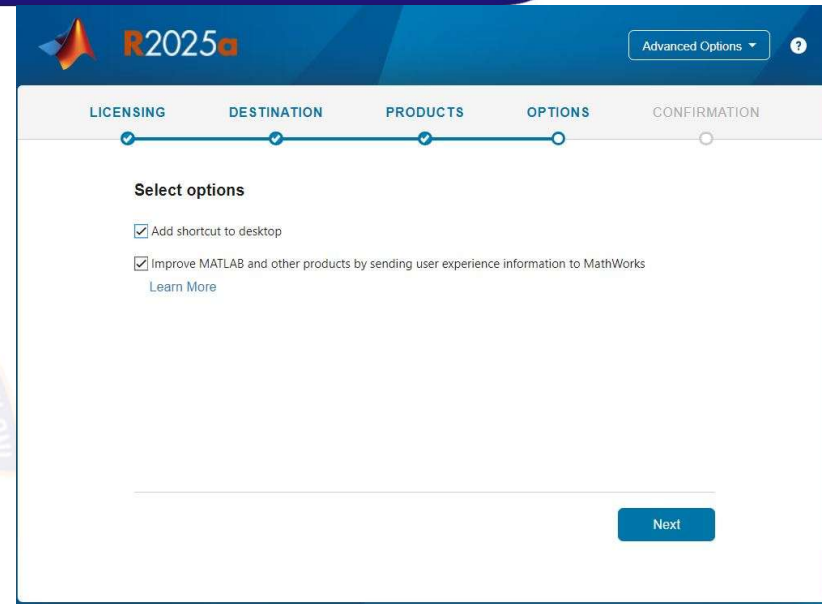
Installing MATLAB

- ❑ Select the destination folder.
- ❑ Then click Next



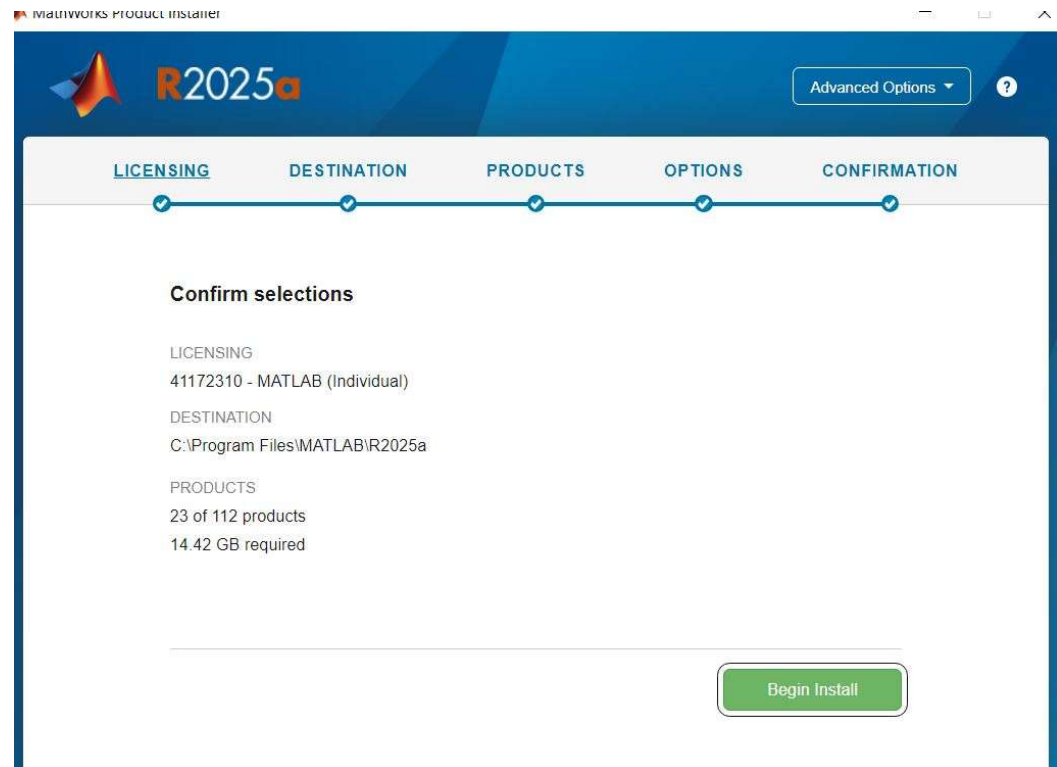
Installing MATLAB

- ❑ Select the required product MATLAB
- ❑ Then click Next
- ❑ In select options, select Add shortcut to desktop
- ❑ Click Next



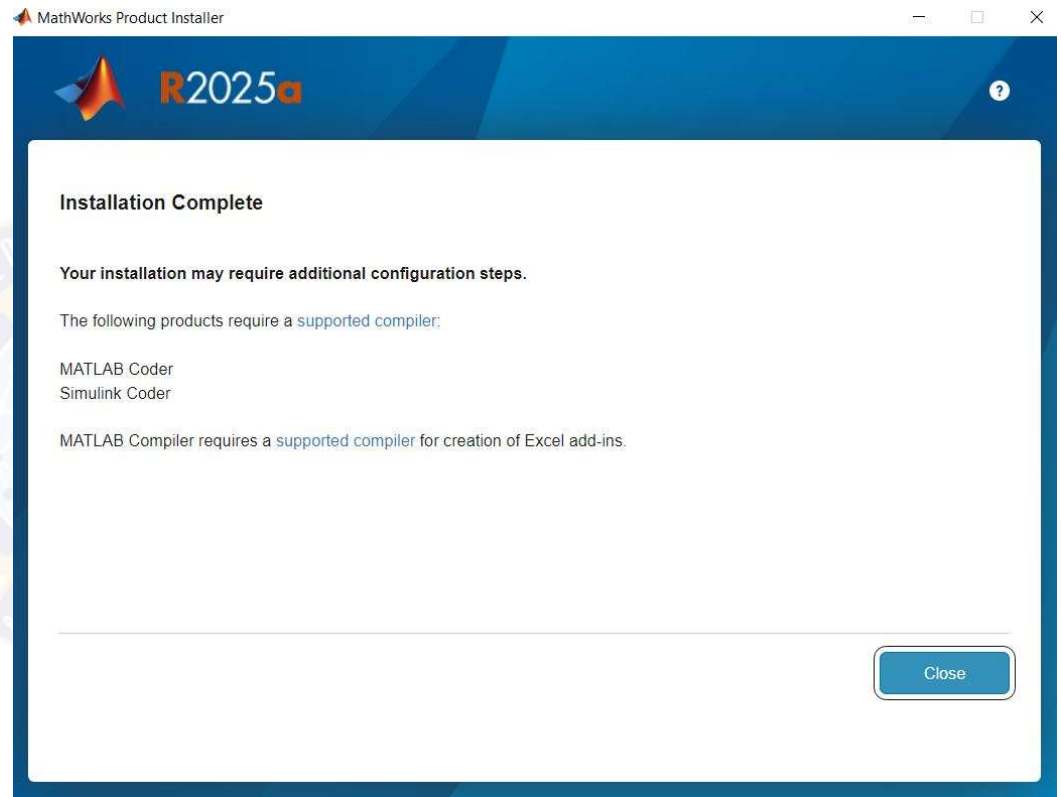
Installing MATLAB

- ❑ Confirm selections and
- ❑ Click Begin Install



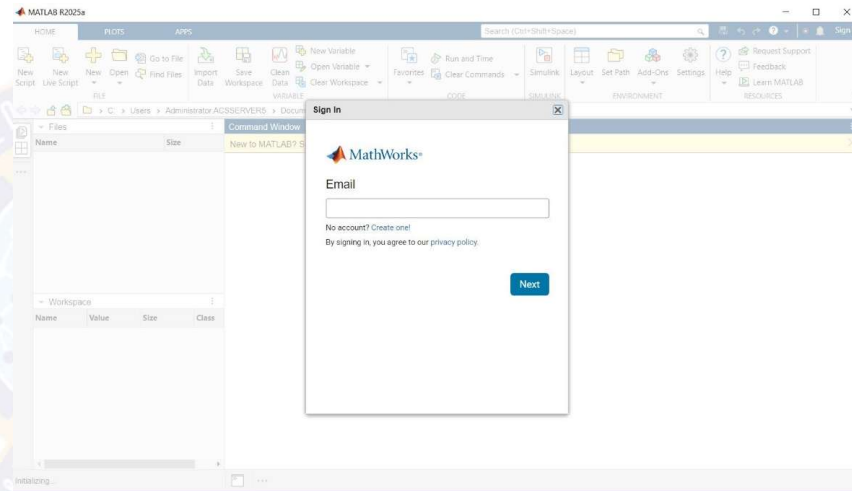
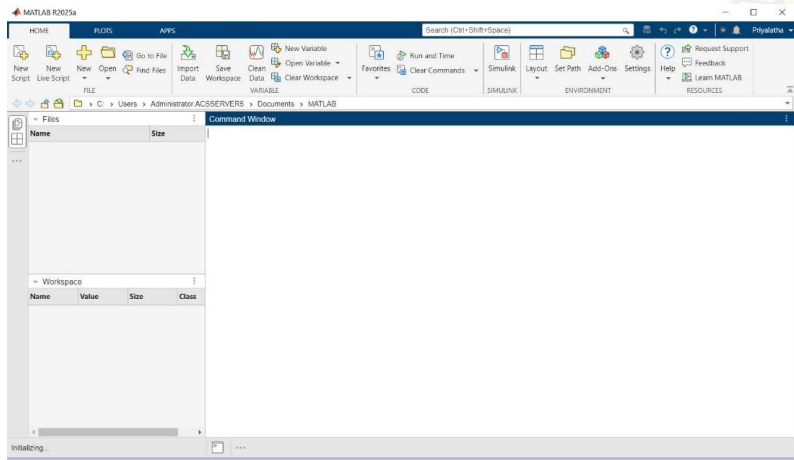
Installing MATLAB

- ❑ Once the installation is done,
close the window



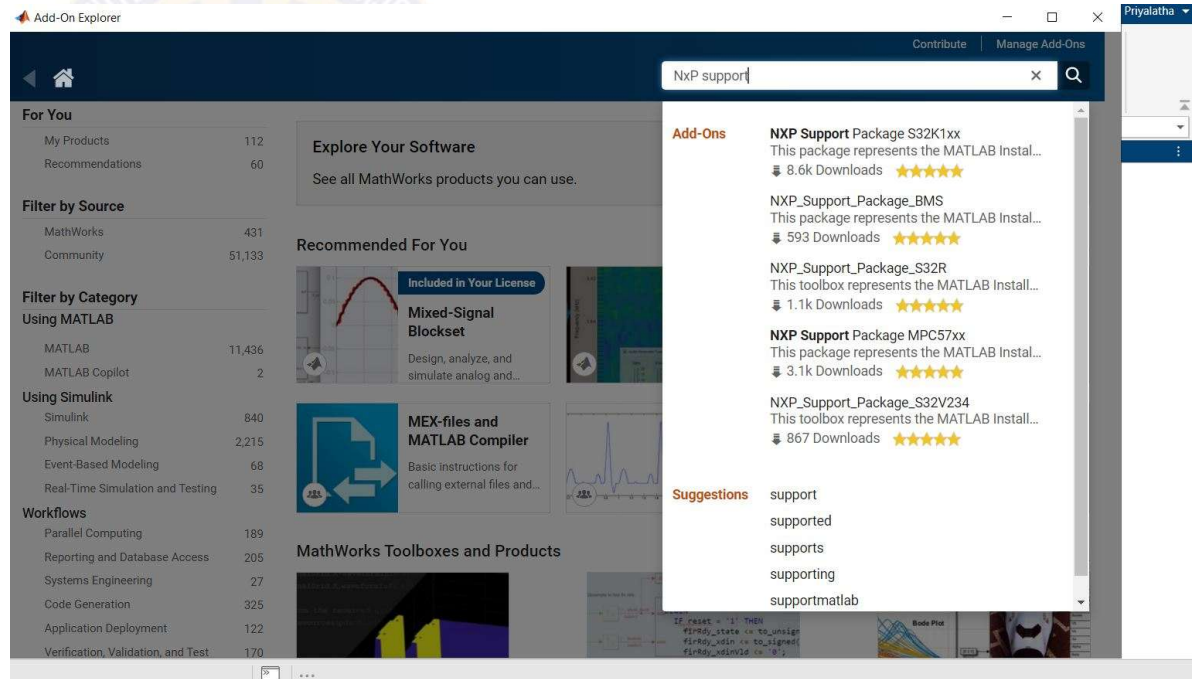
Installing MATLAB

- ❑ Open the MATLAB, by clicking the MATLAB icon.
- ❑ Sign in to the account
- ❑ You can see the command window



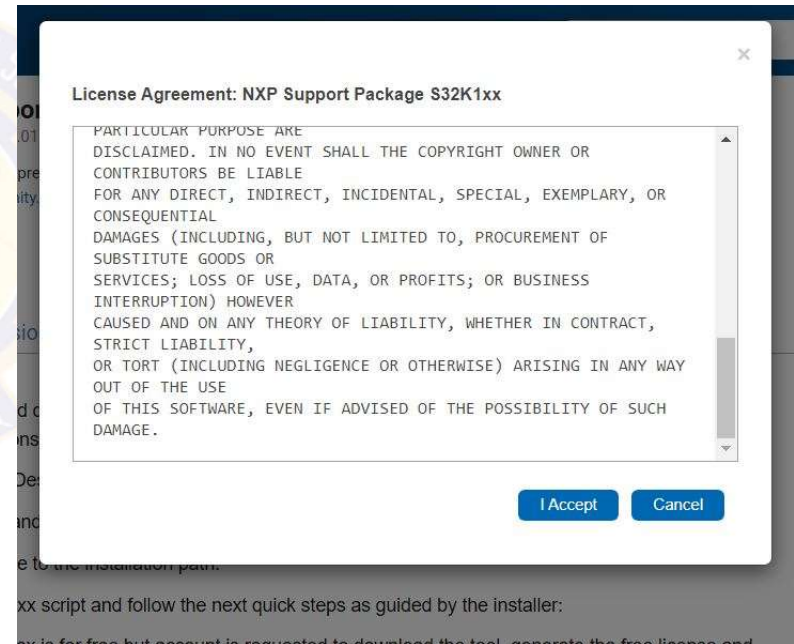
Installing MATLAB

- ❑ Double click Add-ons Explorer
- ❑ Search for NXP Support PACKAGE S32K1xx
- ❑ Start installing the add-on



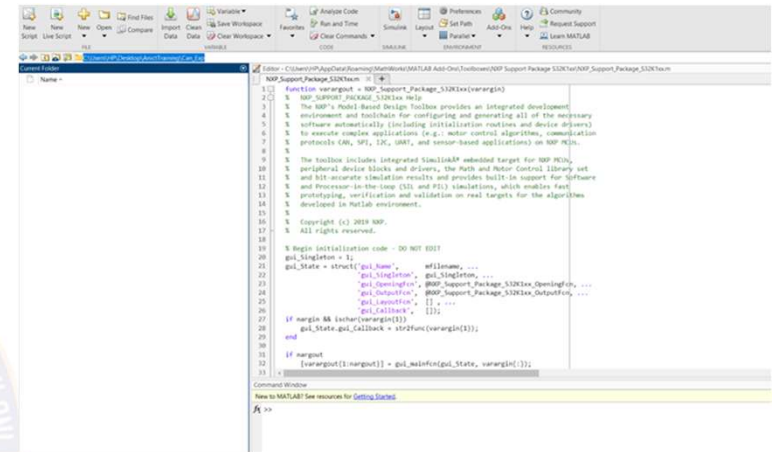
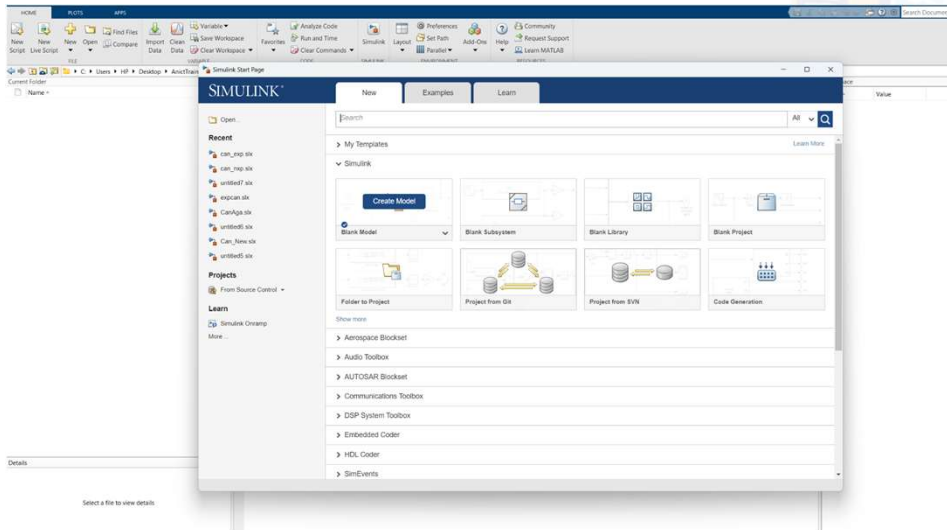
Installing MATLAB

- ❑ After successful completion of installation
- ❑ Click I accept in the license agreement



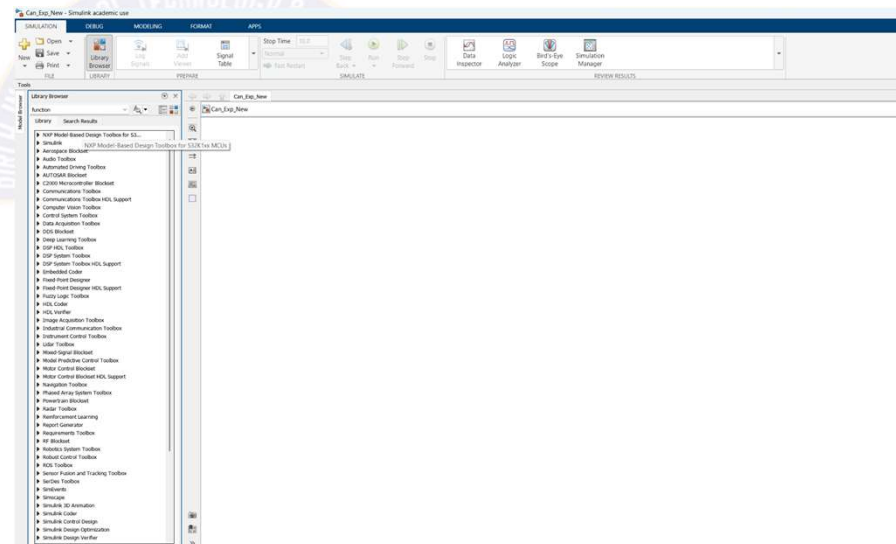
MATLAB Simulink Modelling

- ❑ Open the NXP support package
- ❑ Then create a new Simulink Model
- ❑ Save the Simulink model using your BITS ID



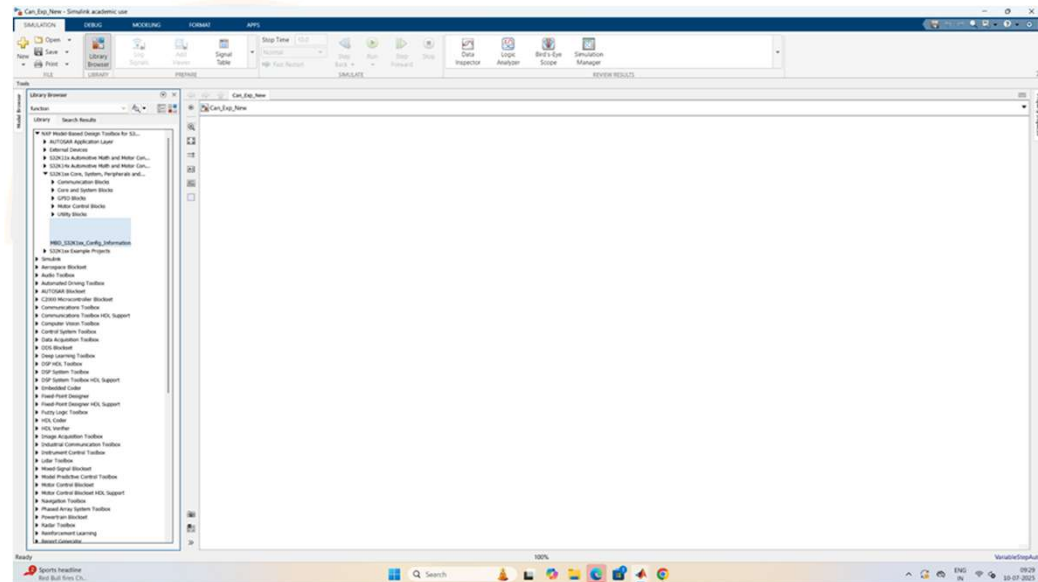
MATLAB Simulink Modelling

- ❑ Go to the library browser and search for “NXP Model design based toolbox” and open it.

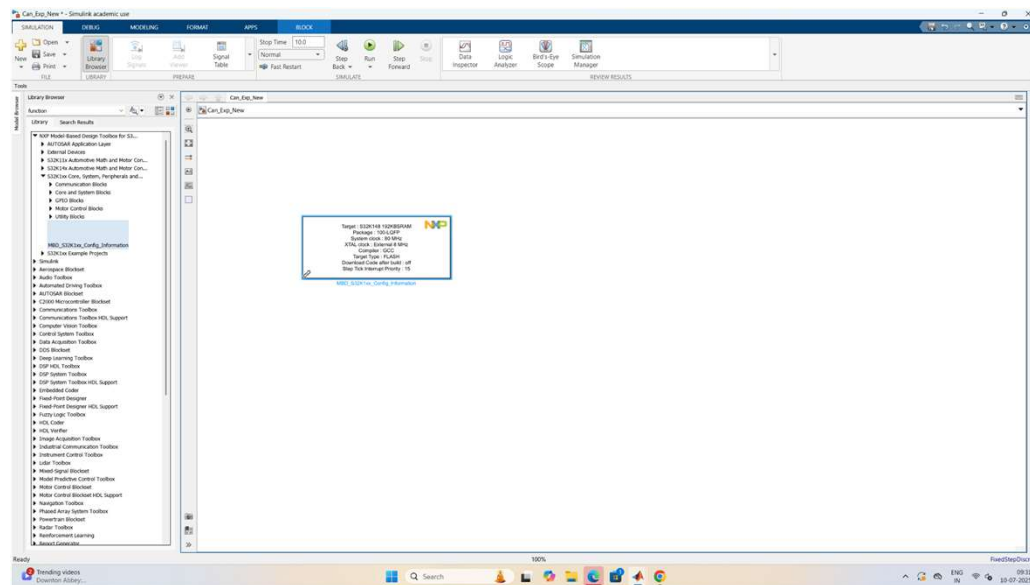


MATLAB Simulink Modelling

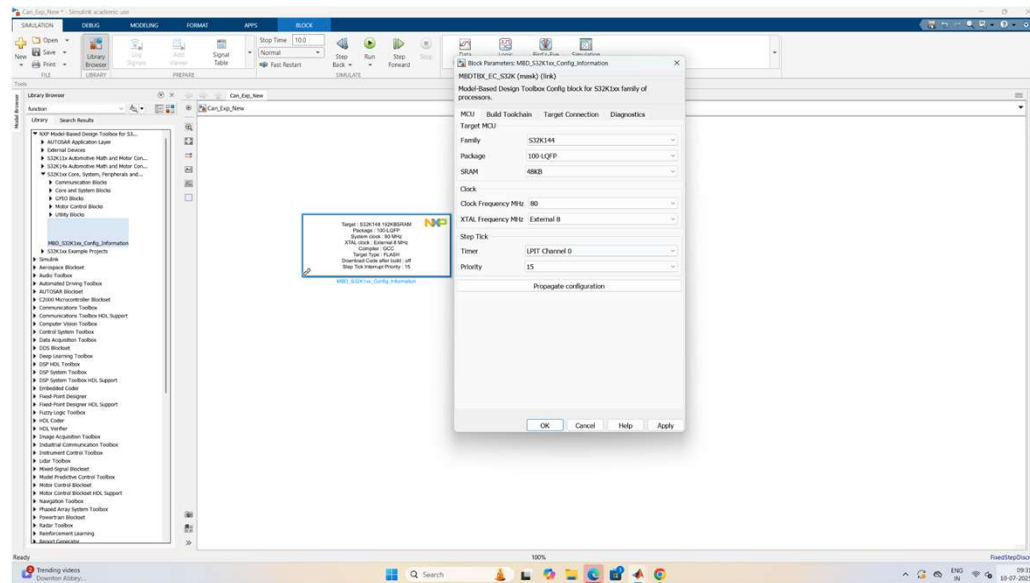
- ❑ Select and Add “The Config Information Block” and configure the following parameters



MATLAB Simulink Modelling

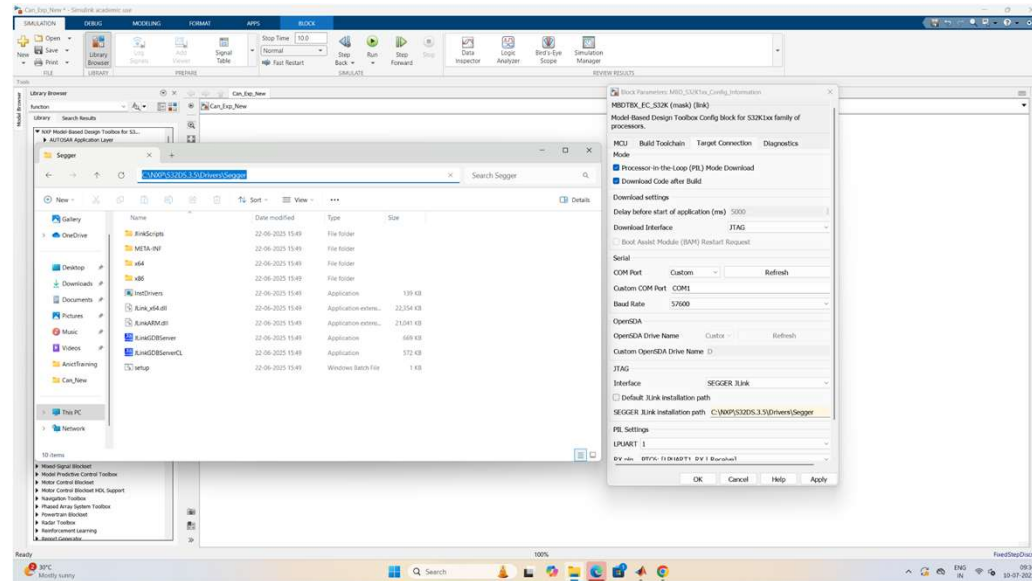


MATLAB Simulink Modelling



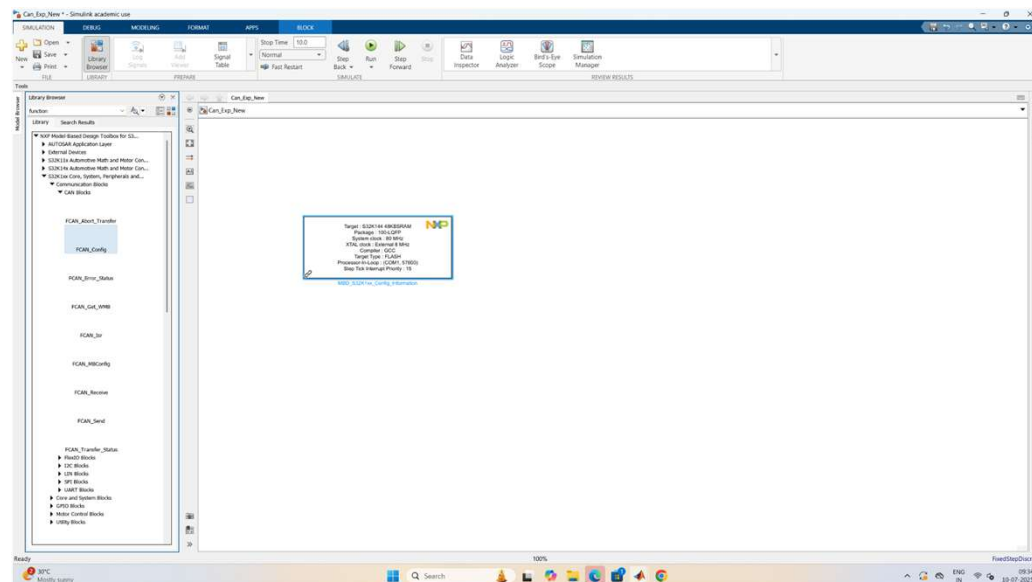
MATLAB Simulink Modelling

❑ Select the Segger JLINK Installation Path



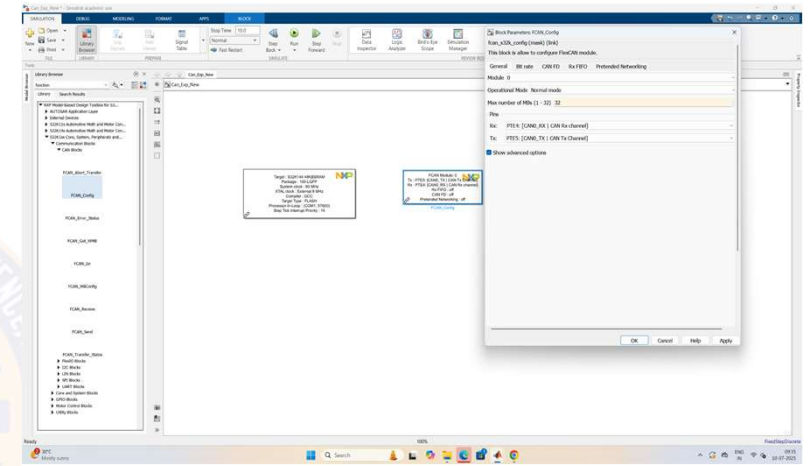
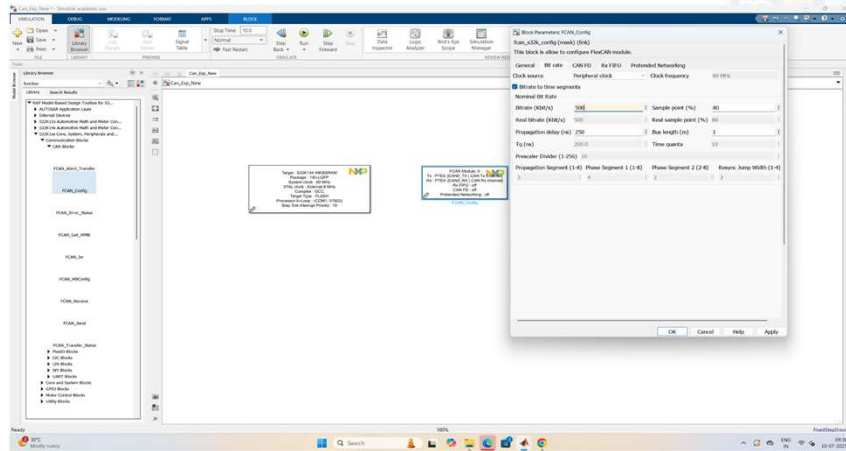
MATLAB Simulink Modelling

□ Add FCAN_Config block



MATLAB Simulink Modelling

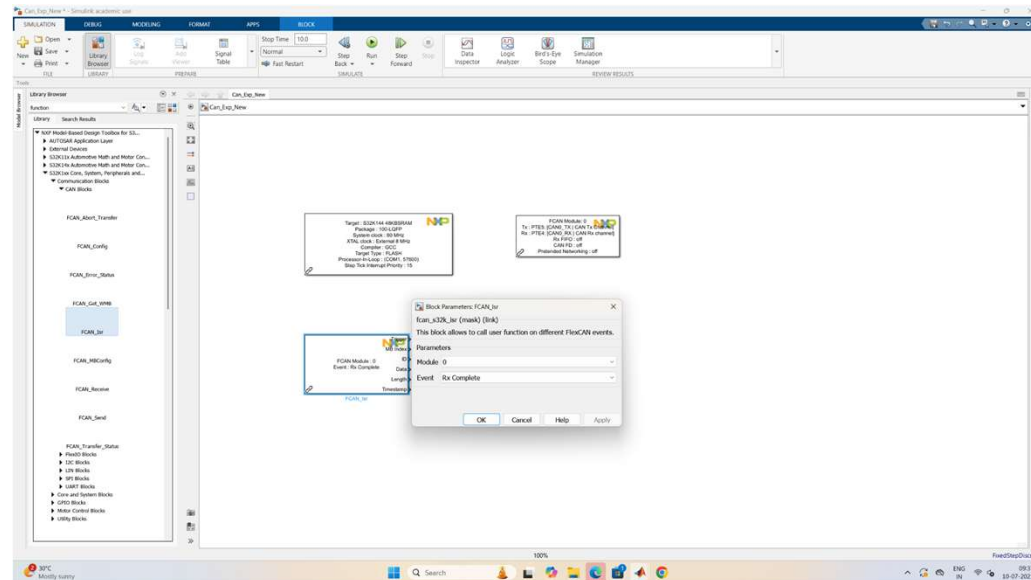
- ❑ **Change the FCAN_Config block parameters**
- ❑ **Select the number of Mail Boxes as 32**
- ❑ **Select the Bit rate as 500 Kbit/s**



❑ Add FCAN_Isr Block

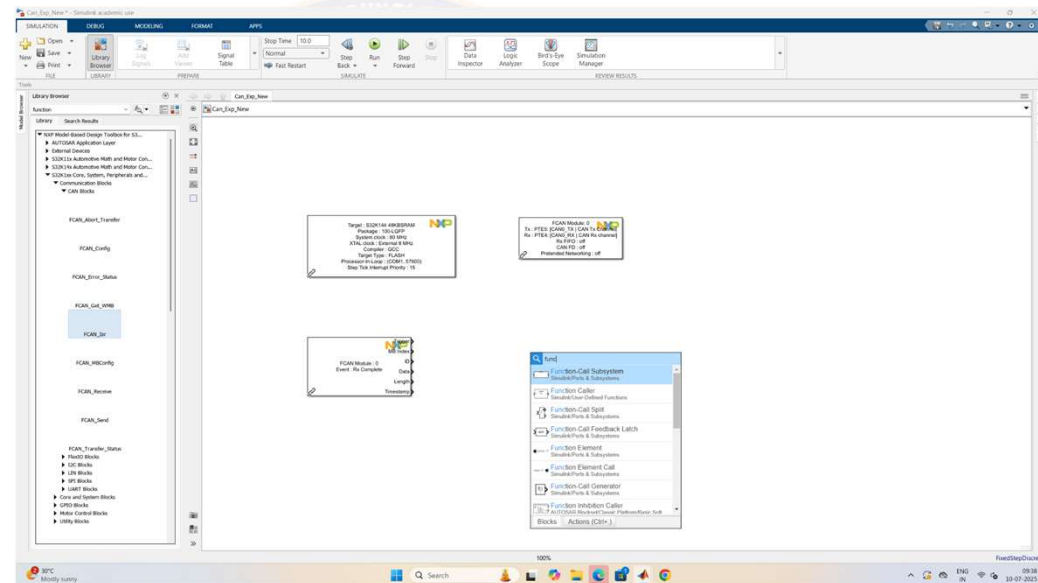
MATLAB Simulink Modelling

- ❑ **Change the block parameters**



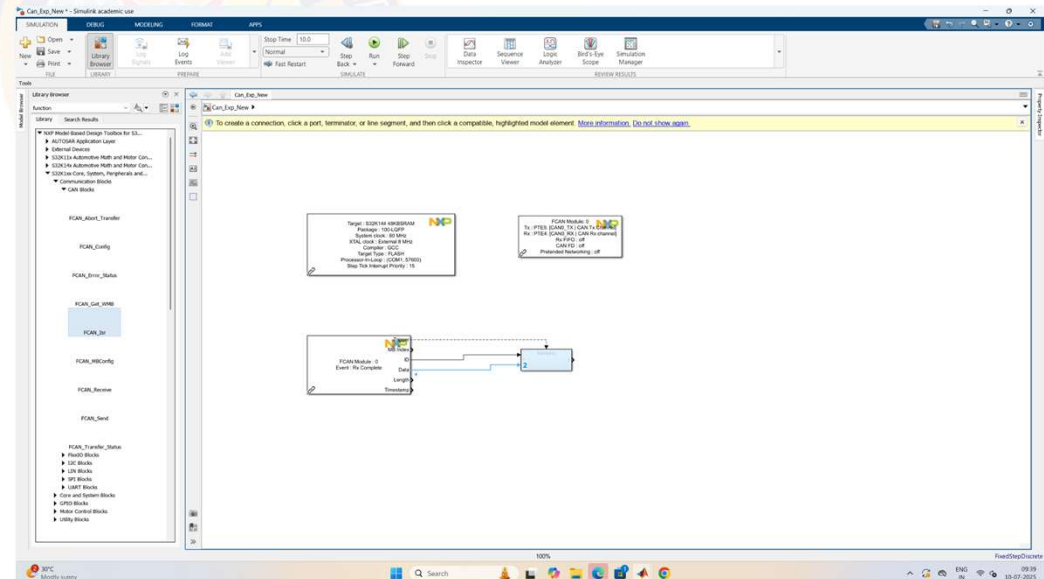
MATLAB Simulink Modelling

❑ Add Function-Call Subsystem Block



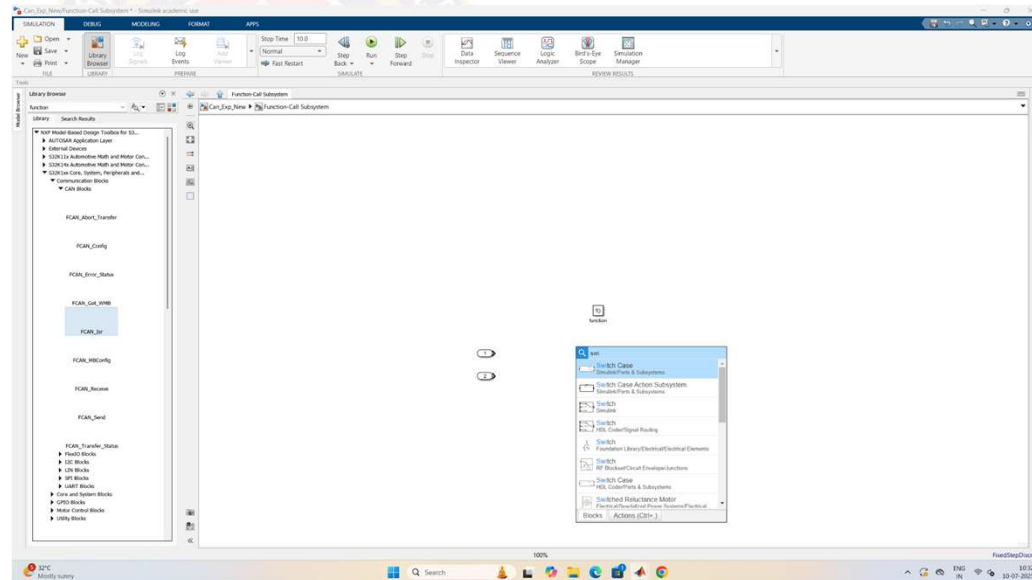
MATLAB Simulink Modelling

- ❑ **Interface the FCAN_Isr and Function-Call Subsystem as Shown below (Add one more port to this block by holding the left click, dragging and dropping till the block)**



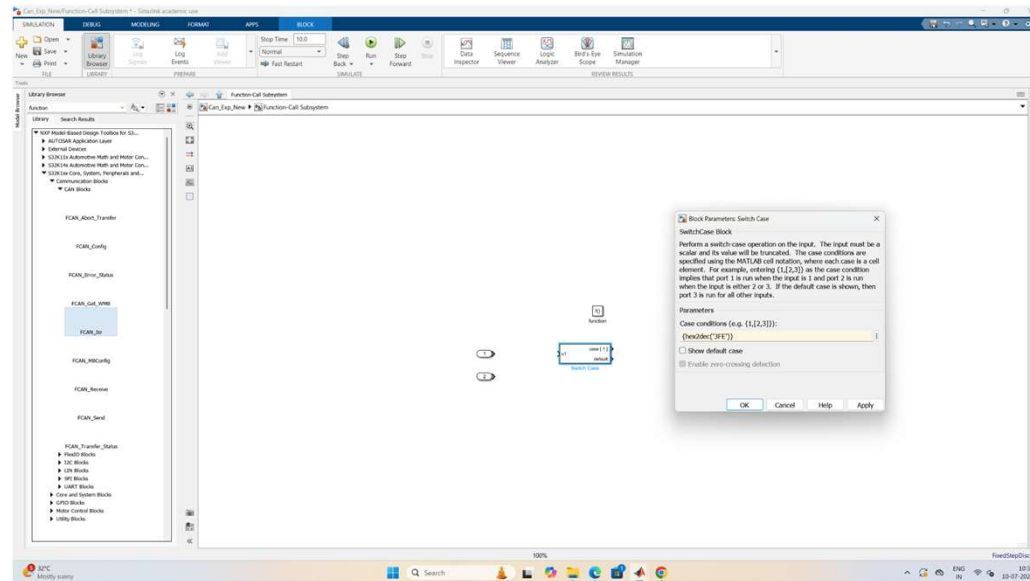
MATLAB Simulink Modelling

- ❑ Inside the above subsystem, add “Switch-Case” block and configure the block as shown below



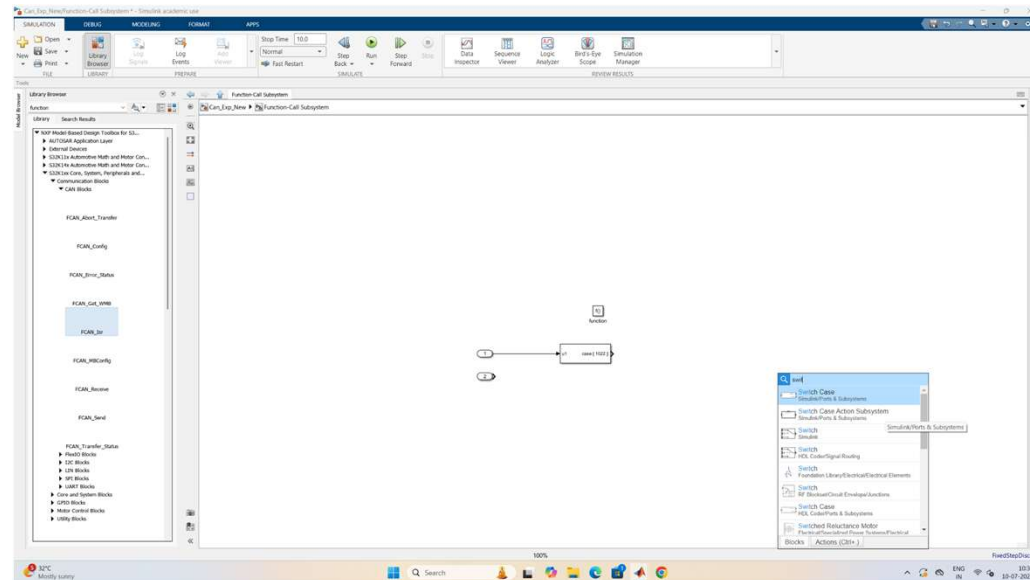
MATLAB Simulink Modelling

❑ Enter CAN ID of the single message



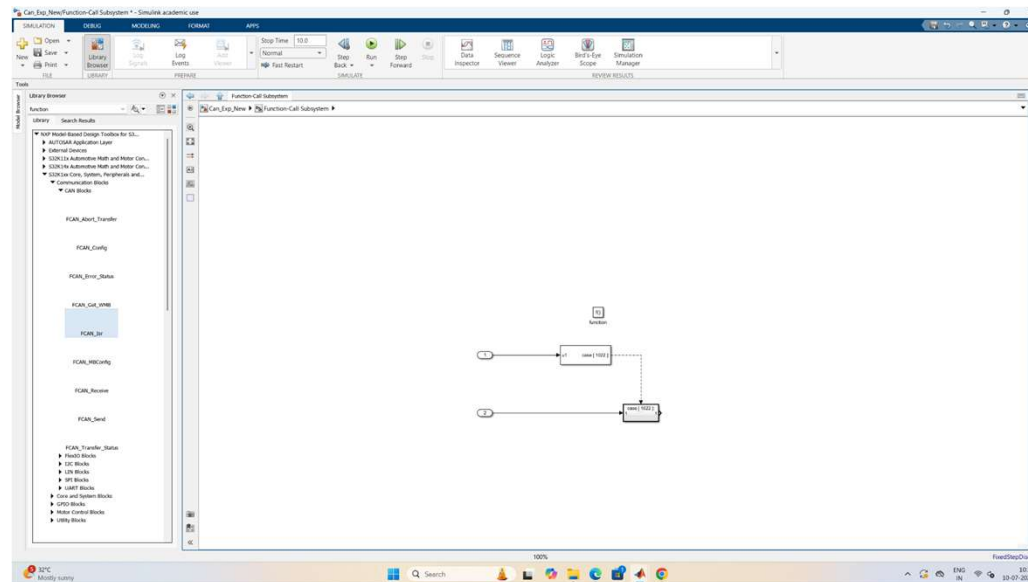
MATLAB Simulink Modelling

- ❑ Add Switch Case Function Block as shown below**



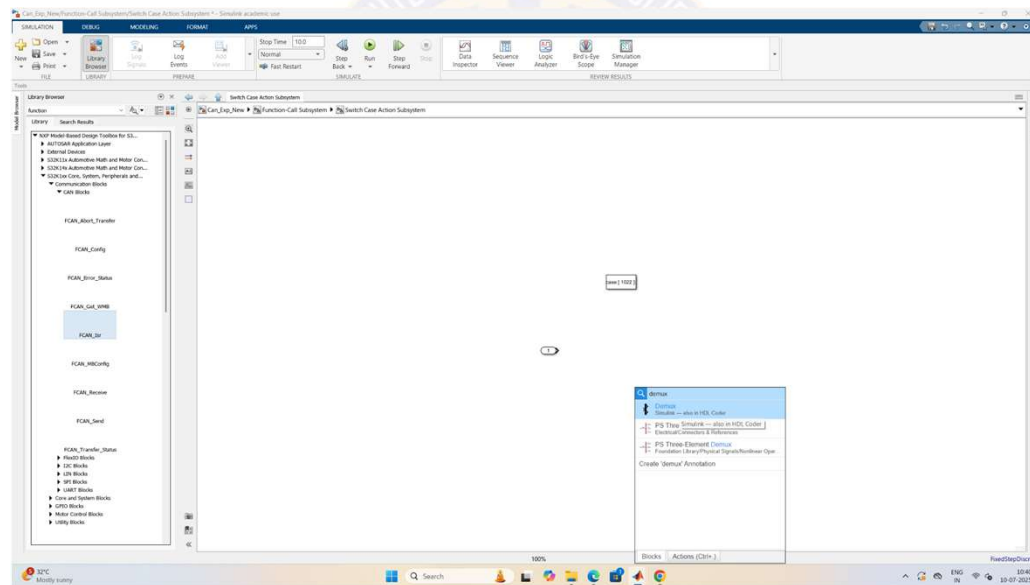
MATLAB Simulink Modelling

- ❑ **Connect the Switch Case Function with other blocks as shown below**



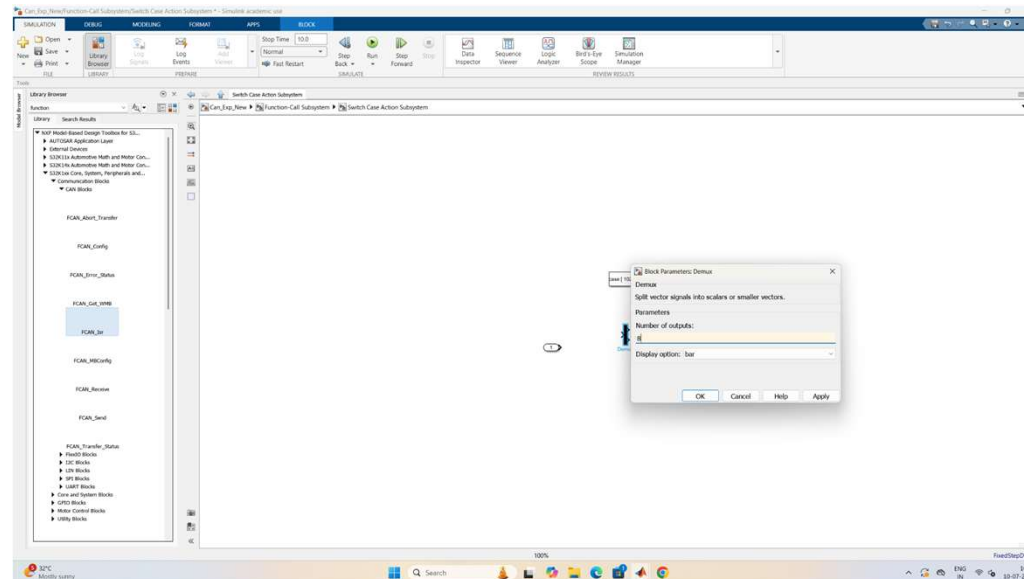
MATLAB Simulink Modelling

- ❑ Inside the Switch Case Function add and connect the following blocks as shown in the figure



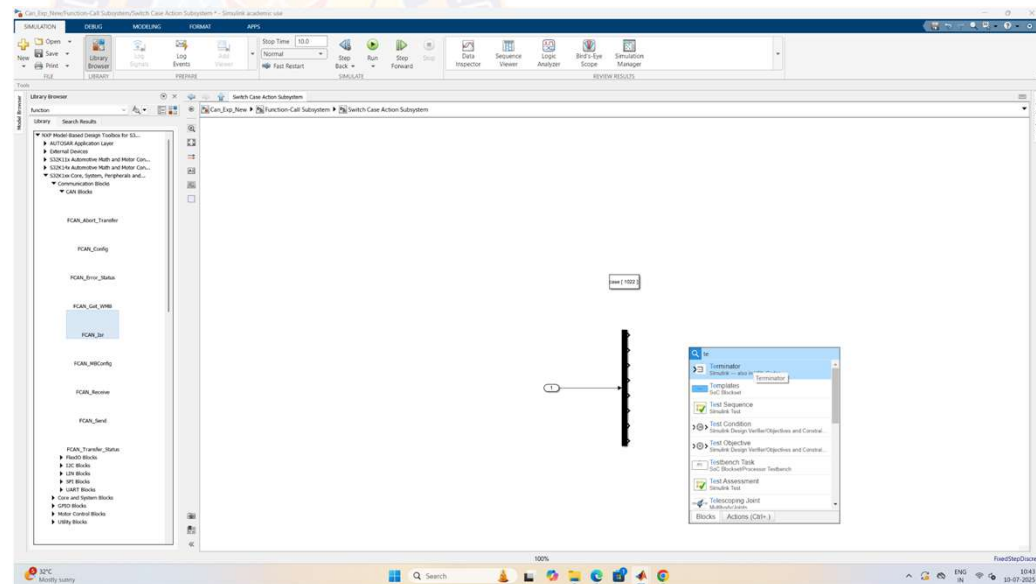
MATLAB Simulink Modelling

- ❑ Select number of inputs to De-Mux as 8 (8 Byte CAN frame)



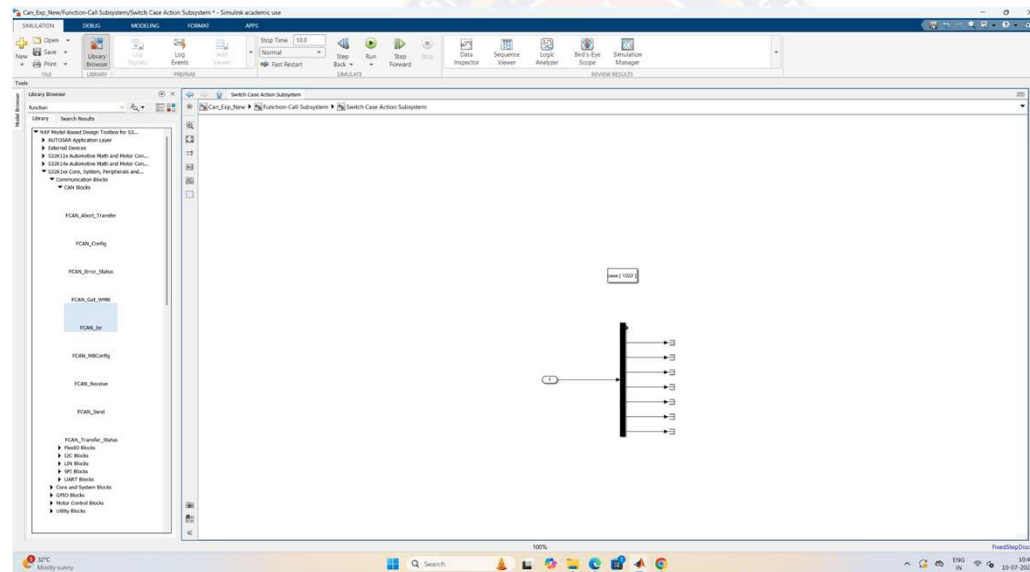
MATLAB Simulink Modelling

- ❑ Connect the input port to input of De-Mux and add the terminator Block
- ❑ All unused bytes from 1-7 should be terminated



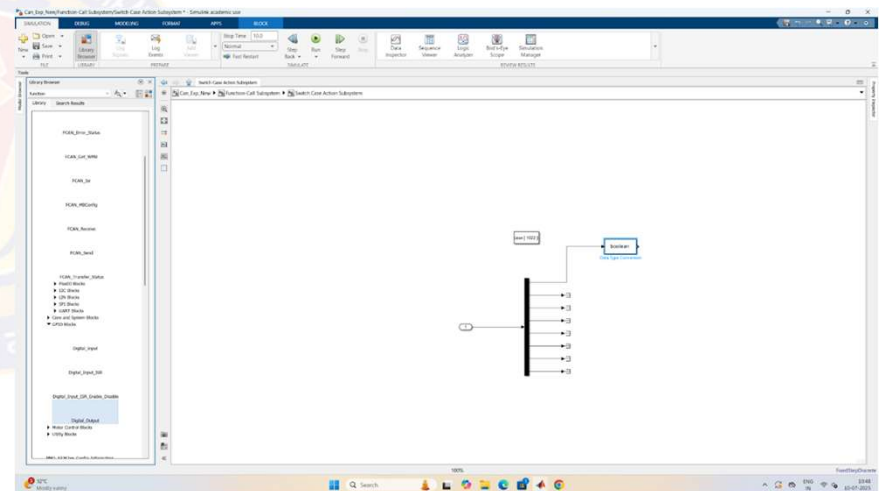
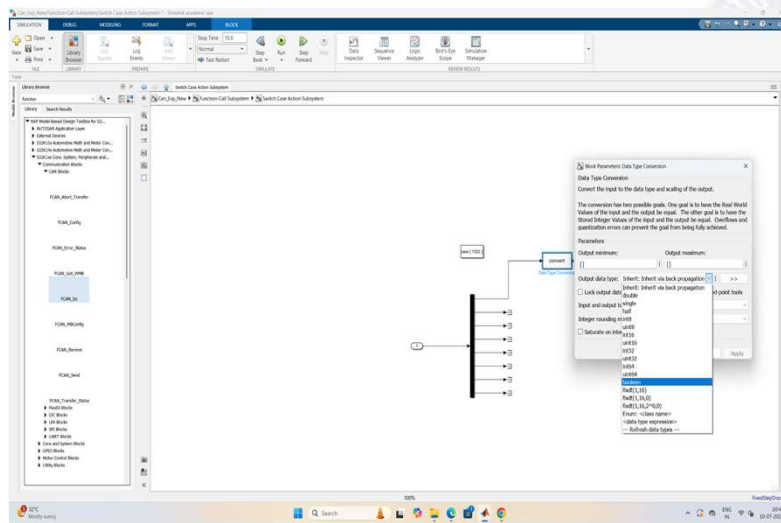
MATLAB Simulink Modelling

- ❑ **Select Data-Type Conversion Block. Changing the Data Type of Byte 0 to Logical Value**



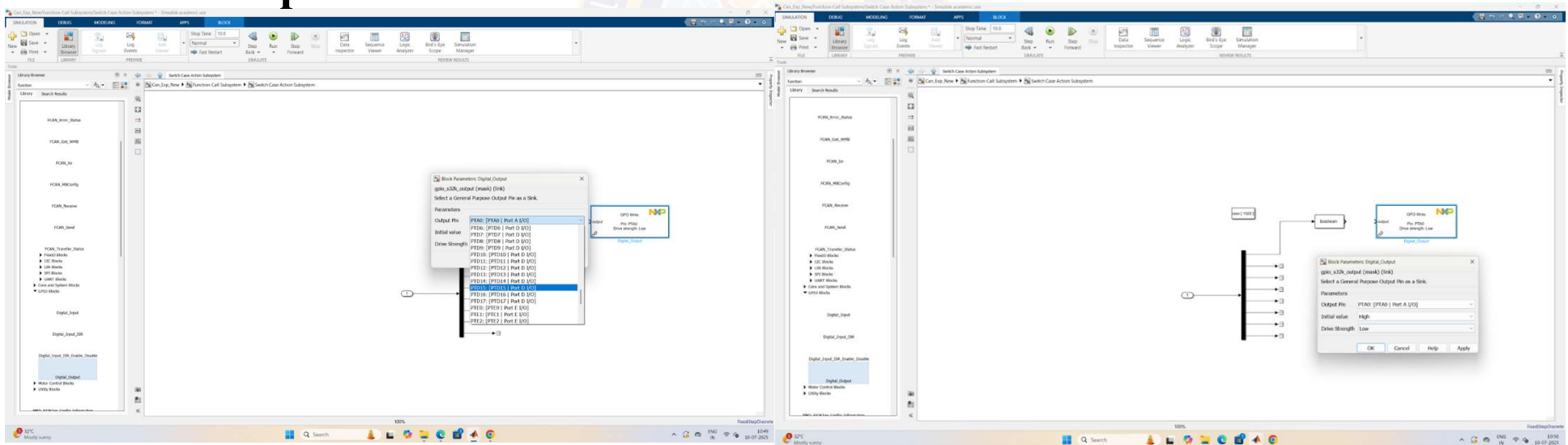
MATLAB Simulink Modelling

- ❑ Output datatype of the Data-Type Conversion Block is changed to Boolean



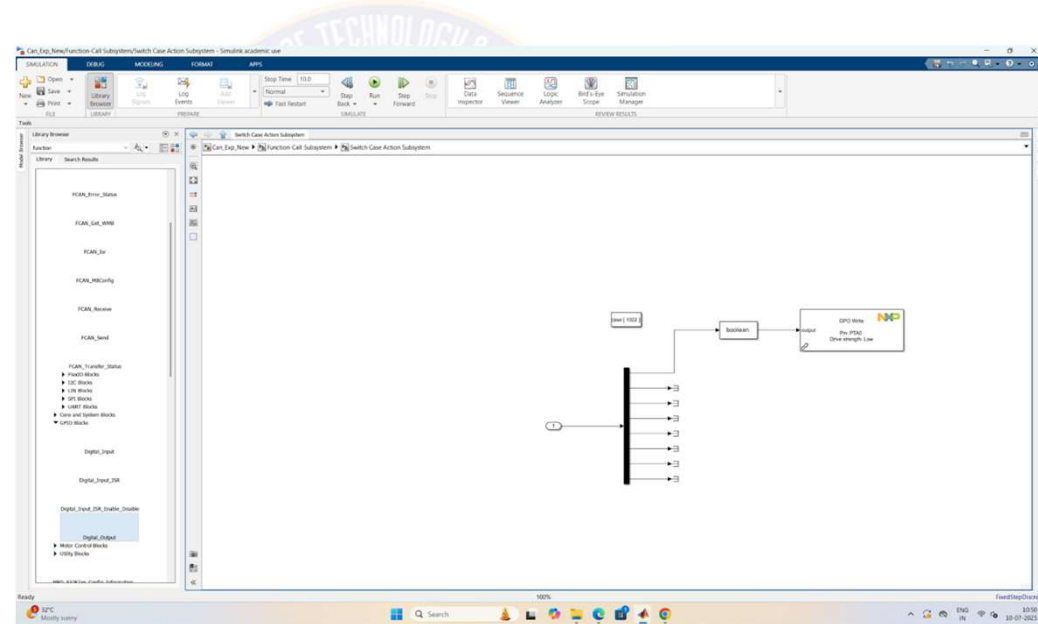
MATLAB Simulink Modelling

- ❑ Add the digital output block
- ❑ Change the digital output parameters.
- ❑ Select PTA7 pin to control Red LED.



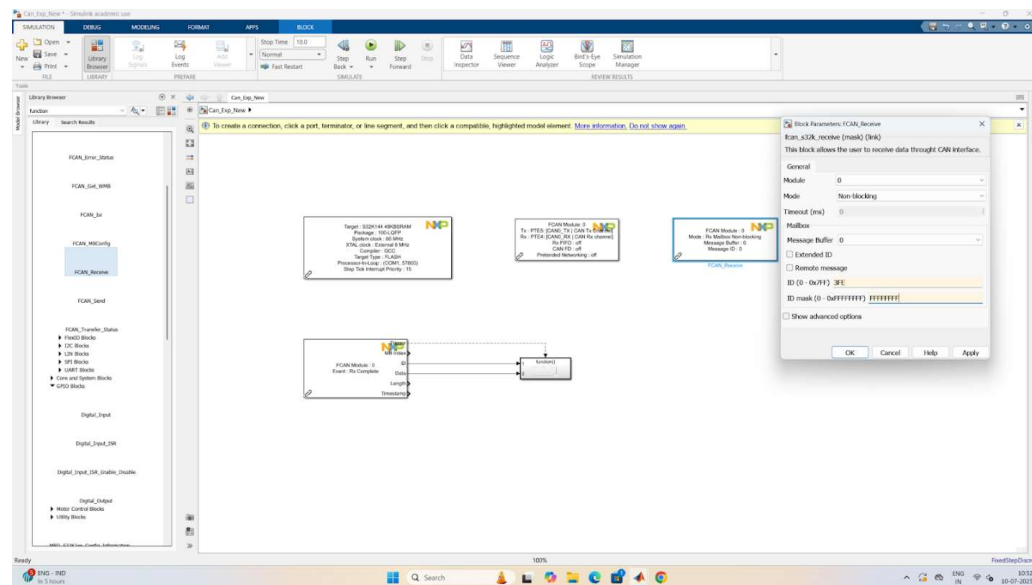
MATLAB Simulink Modelling

□ Final Connection



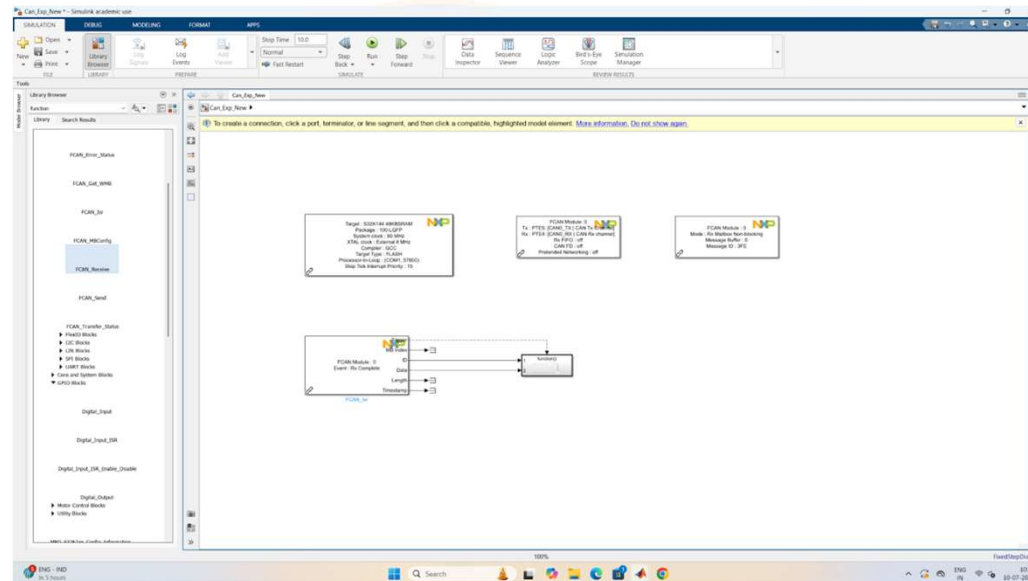
MATLAB Simulink Modelling

- ❑ Mail Box Message Buffer ID is selected as 0 and its corresponding CAN ID is set as 3FE.



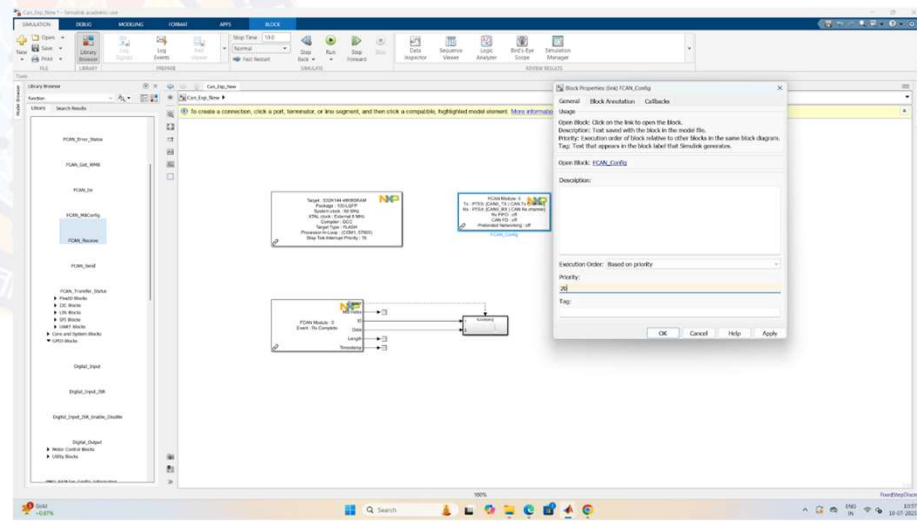
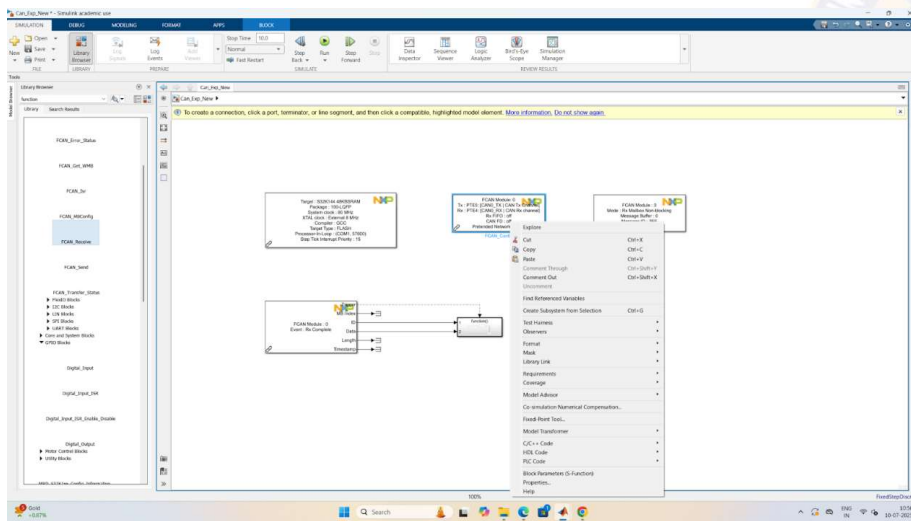
MATLAB Simulink Modelling

- ❑ **Terminate the MB Index, Length and Time Stamp**



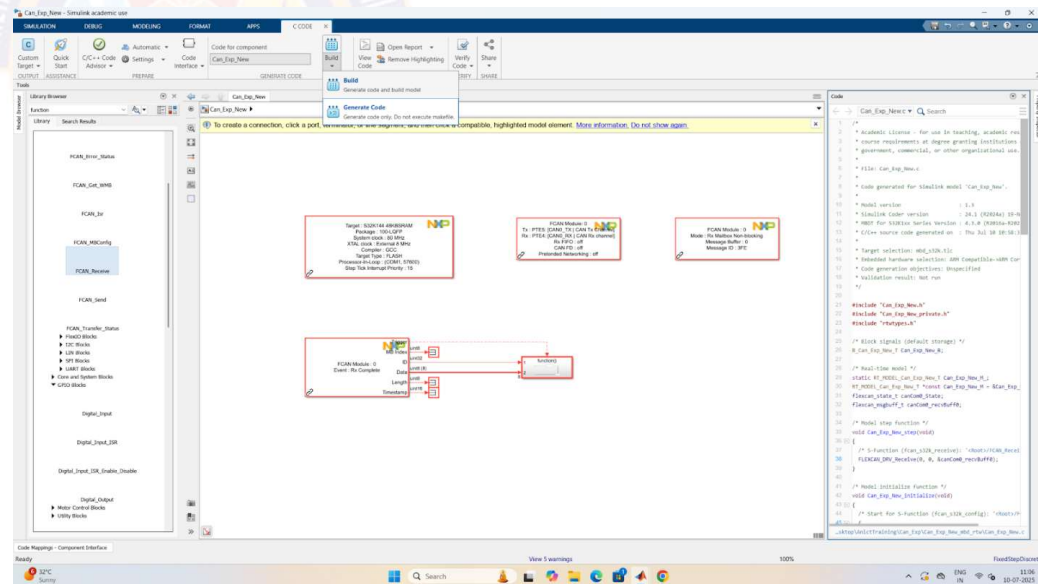
MATLAB Simulink Modelling

- ❑ Set the block properties of FCAN_Config block as shown below



MATLAB Simulink Modelling

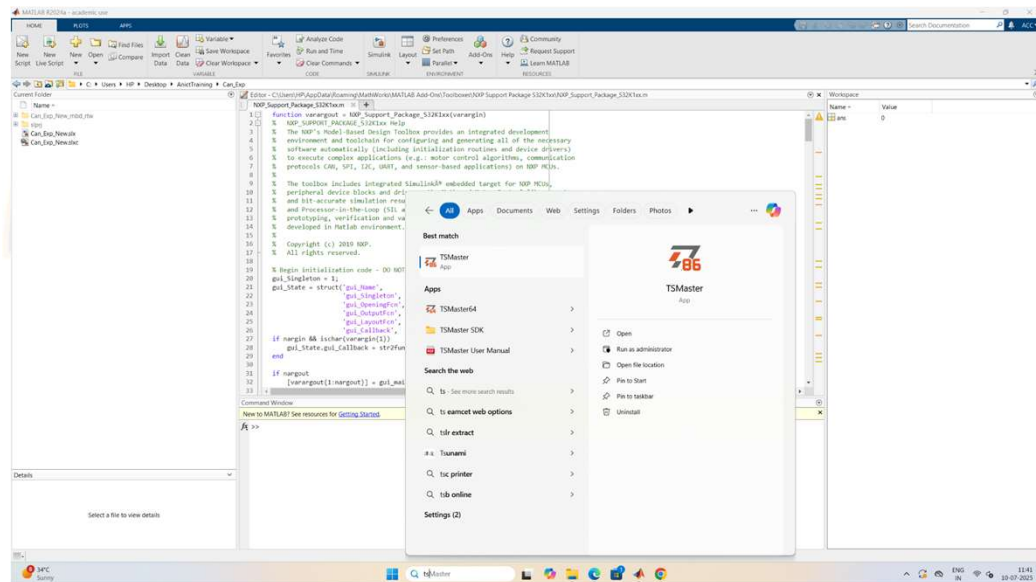
- ❑ Connect the target board. Generate the Code by using Ctrl+D and Ctrl+B or select the option as shown below build and generate code.
- ❑ You can see the generated code only when there is no error. If not, look for errors and rectify the same.



TS MASTER

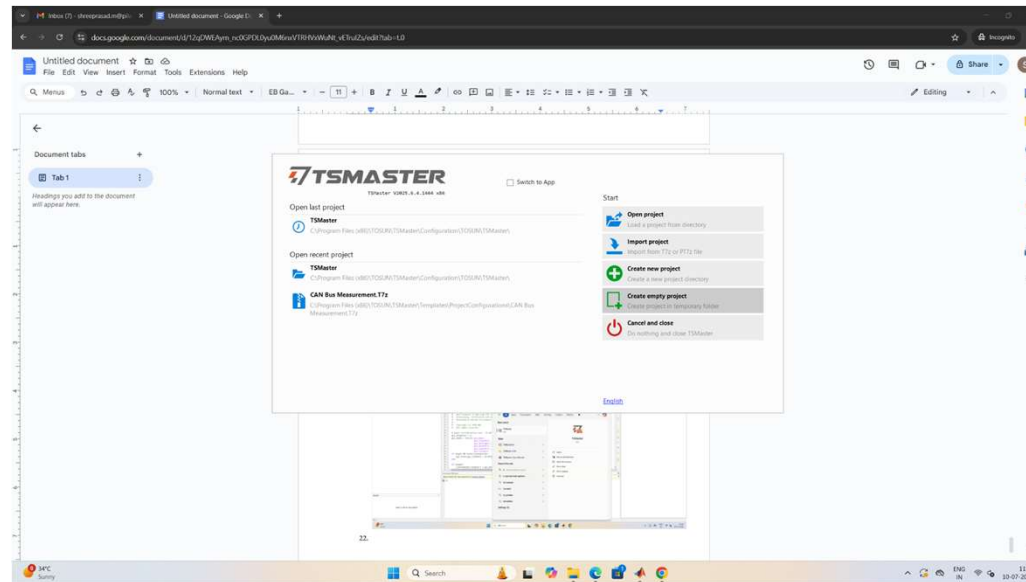
❑ Download TS Master Application

❑ Open it



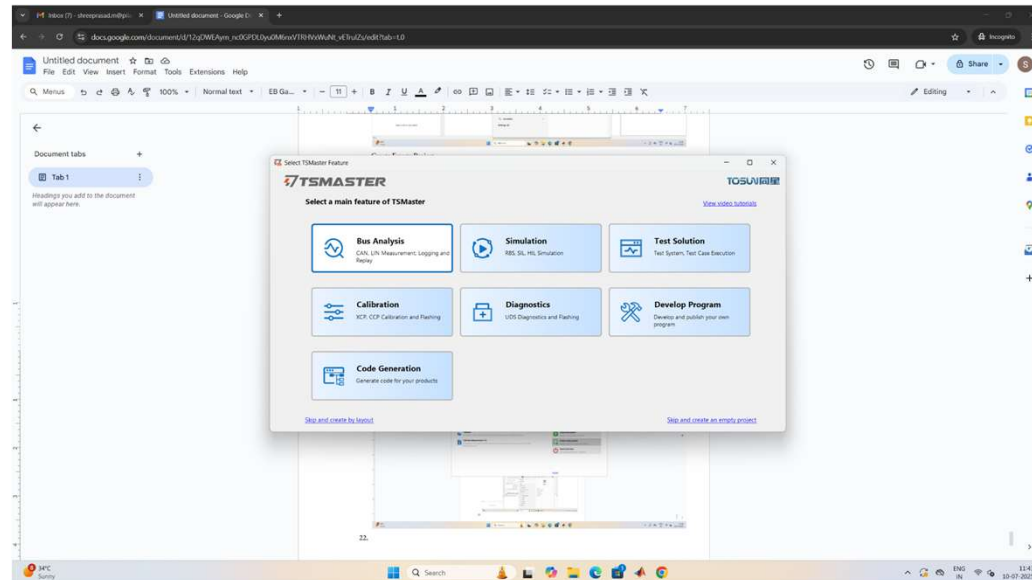
TS MASTER

❑ Create new project



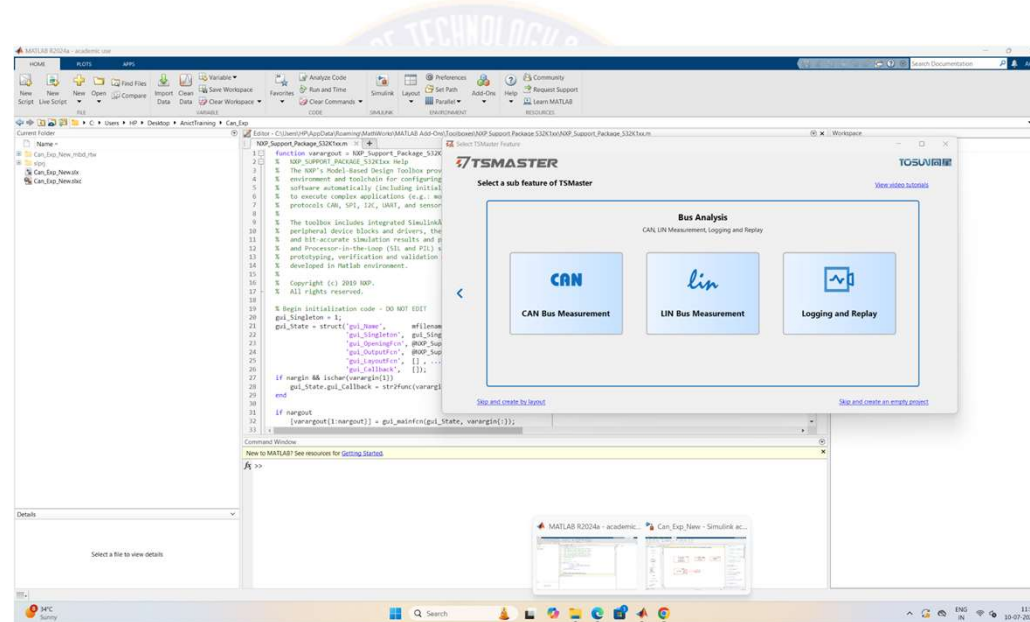
TS MASTER

❑ Select Bus Analysis



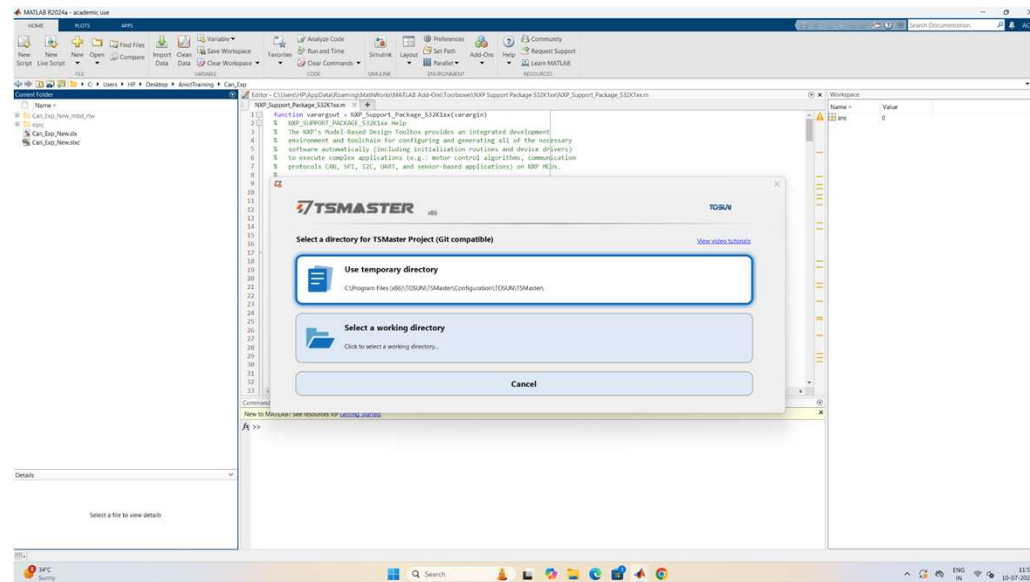
TS MASTER

❑ Select CAN bus measurement



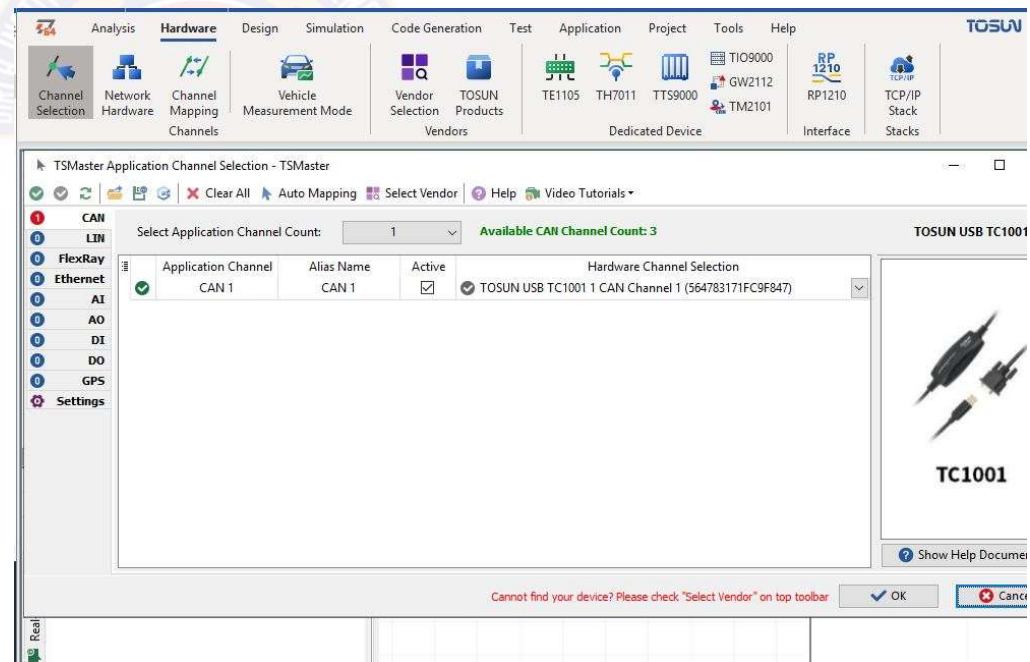
TS MASTER

❑ Select temporary directory



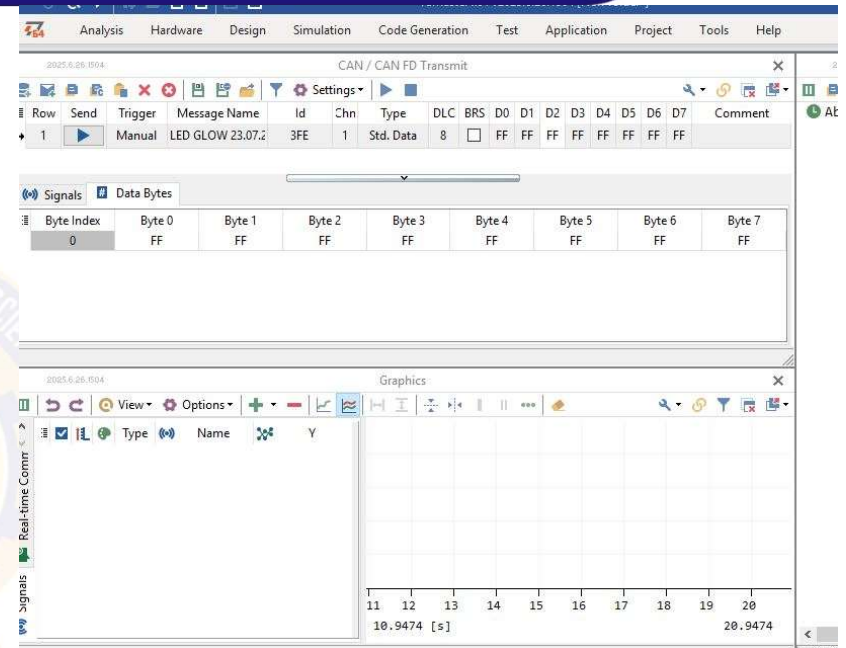
TS MASTER

- ❑ Click Channel selection
- ❑ Select TOSUN USB TC1001 CAN Channel
- ❑ Click OK



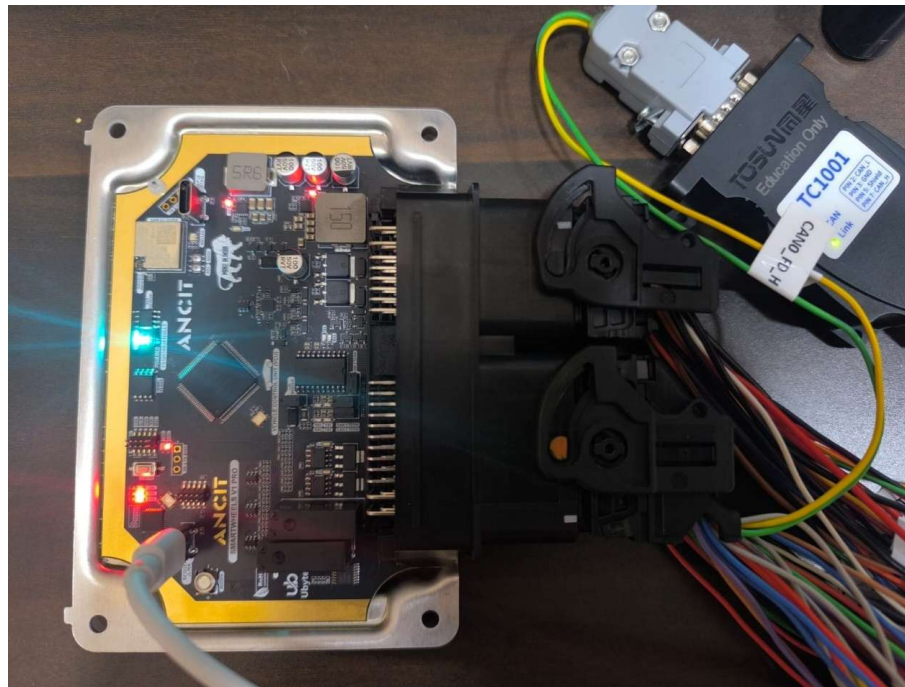
TS MASTER

- ❑ Click Transmit
- ❑ CAN/CANFD Transmit
- ❑ Add a new message
- ❑ Give a message name as LED_your BITS ID
- ❑ Set all the message Bytes as FF (D0 to D7)
- ❑ Enter the values for Bytes (FF – OFF and 00-ON). Relevant is only for D0, and others are redundant. Enter the CAN ID as 3FE and Transmit the frame by clicking Play Button



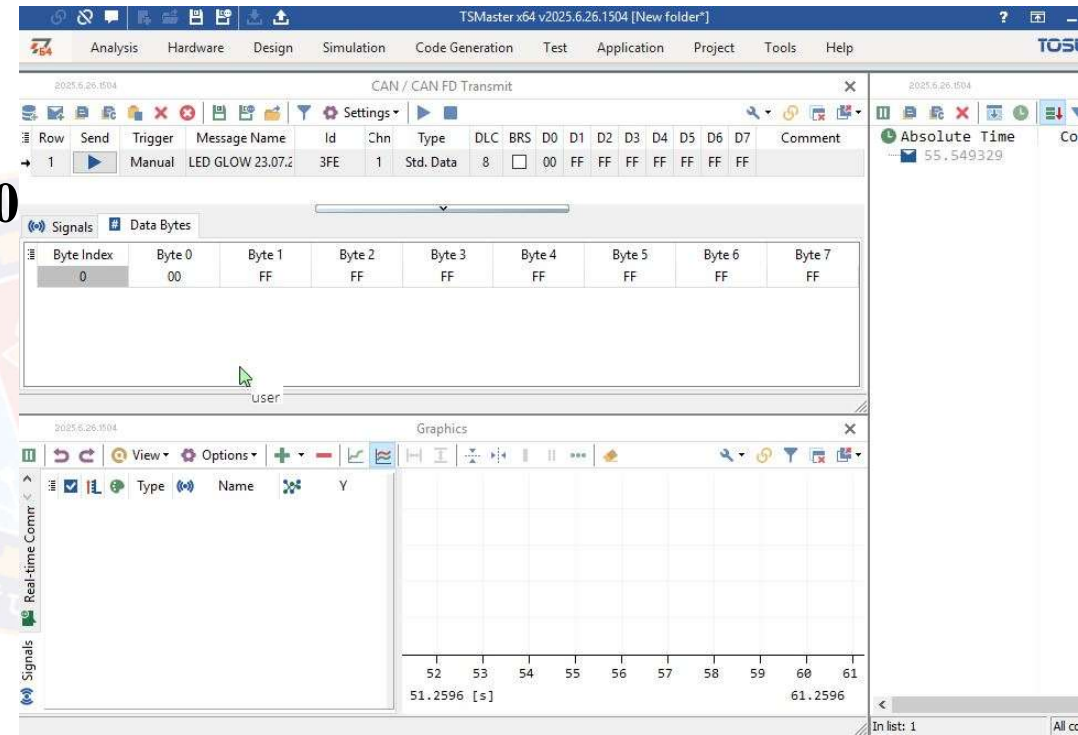
Hardware setup

❑ Smart wheels V1 Pro with Tosun TC1001-CAN connector



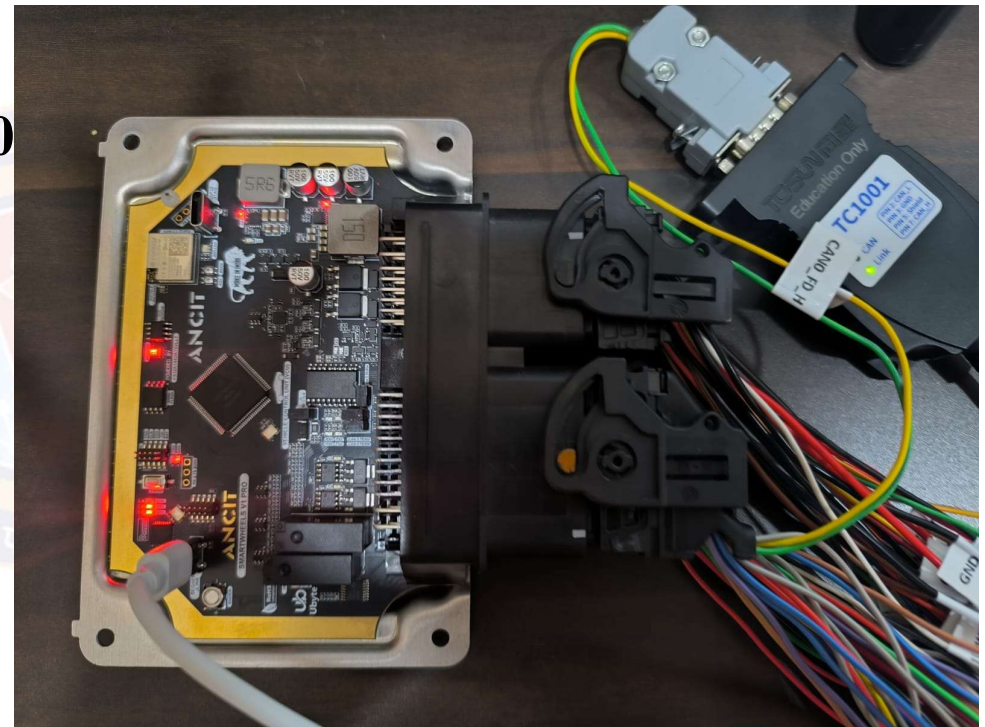
Results

- ❑ CAN byte D0 as 00 (Red LED)
- ❑ Set the D0 byte of CAN message as 00



Results

- ❑ CAN byte D0 as 00 (Red LED)
- ❑ Set the D0 byte of CAN message as 00





Exercise-2

Exercise - 2

Changing the CAN messages Byte by Byte and changing the LED colour.

Aim: To change the CAN messages Byte by Byte and to change the LED colour.

Hardware Requirements: Smart wheels V1 Pro with TC1001.

Software Requirements: MATLAB Simulink with NXP support package S32K1xx

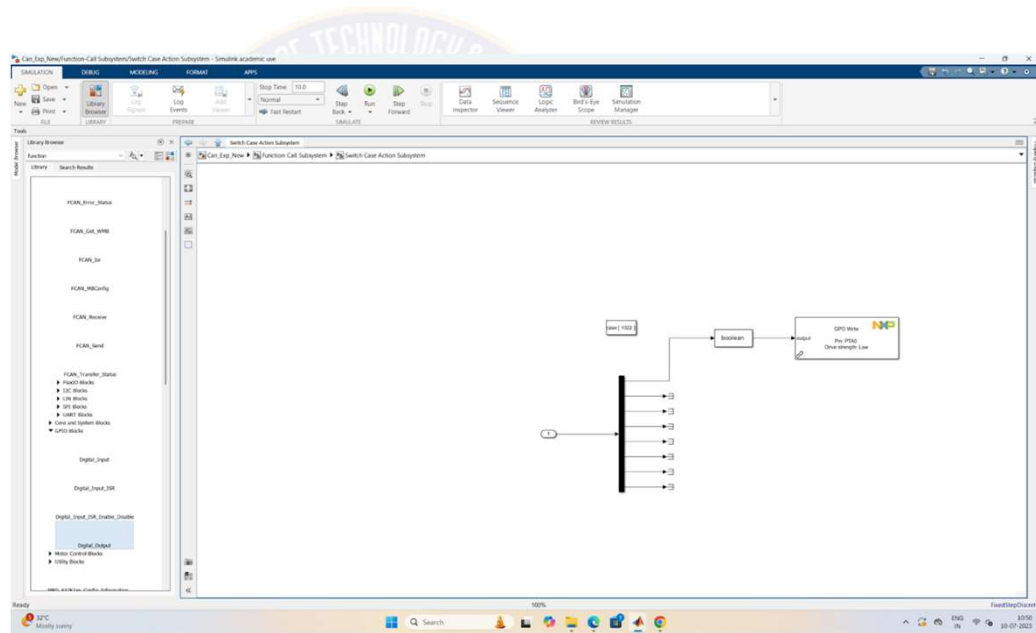
MATLAB Simulink Modelling

- ❑ **Create the switch case action subsystem as shown in the figure.**
- ❑ **PTA7 for red LED (D0)**
- ❑ **PTA6 for green LED (D1) and**
- ❑ **PTE7 for blue LED (D2)**
- ❑ **Terminate all other bytes using terminators**



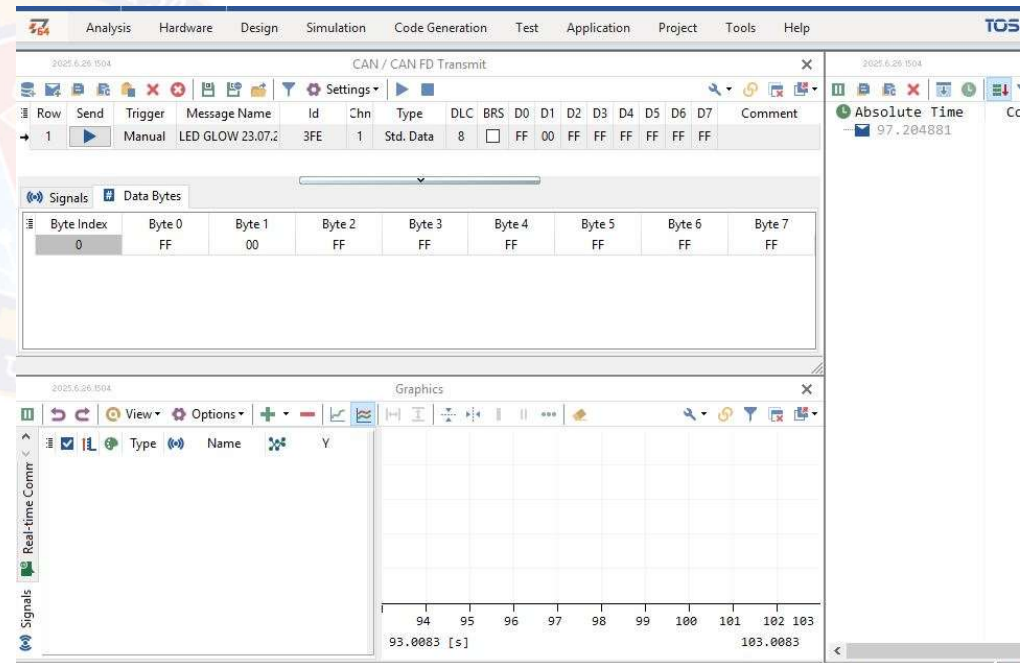
MATLAB Simulink Modelling

□ Final Connection

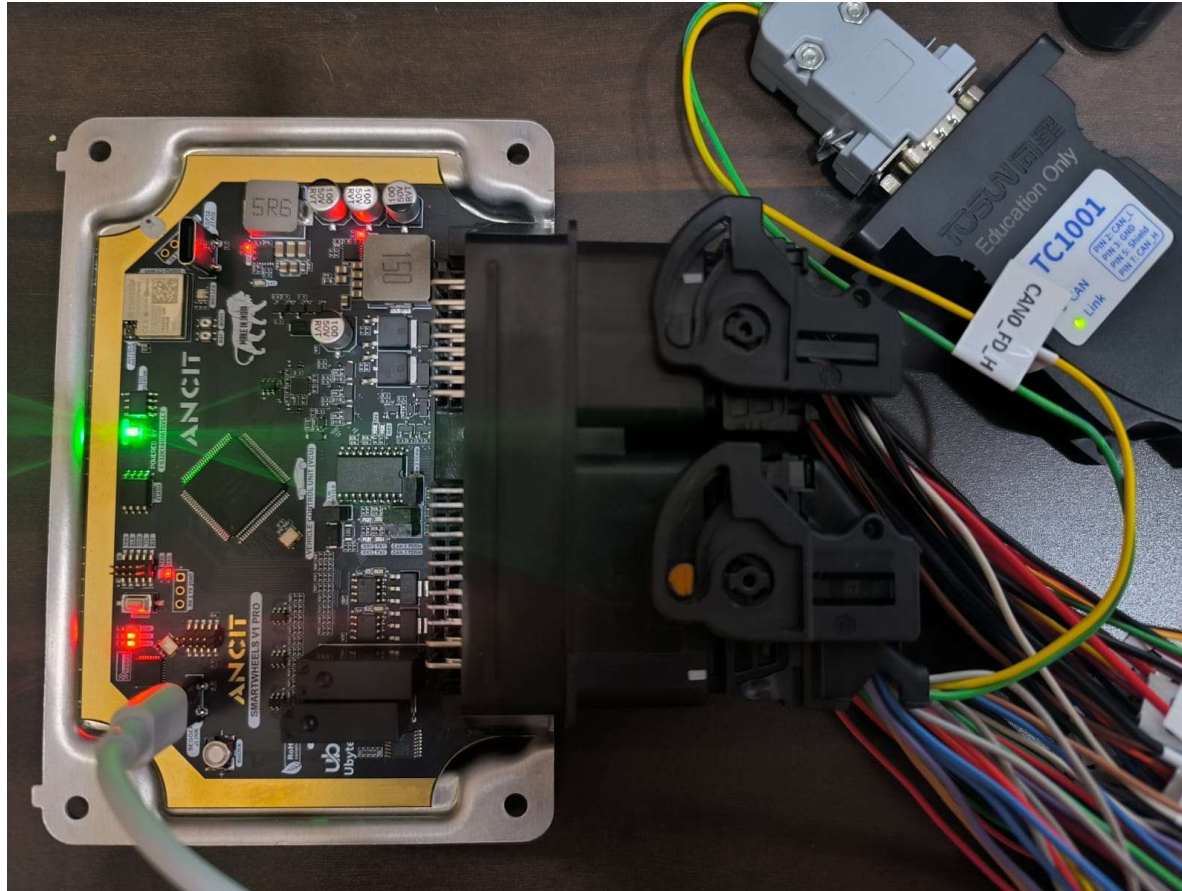


Results

- ❑ CAN byte D1 as 00 (Green LED)
- ❑ Set the D0 byte of CAN message as FF
- ❑ And D1 byte of CAN message as 00

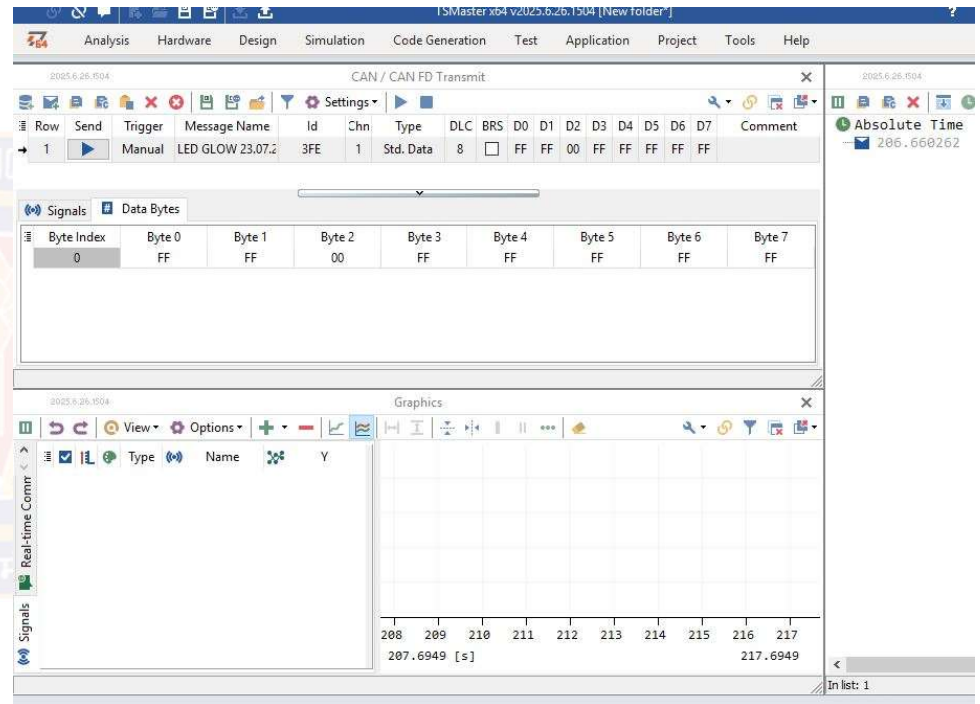


Results

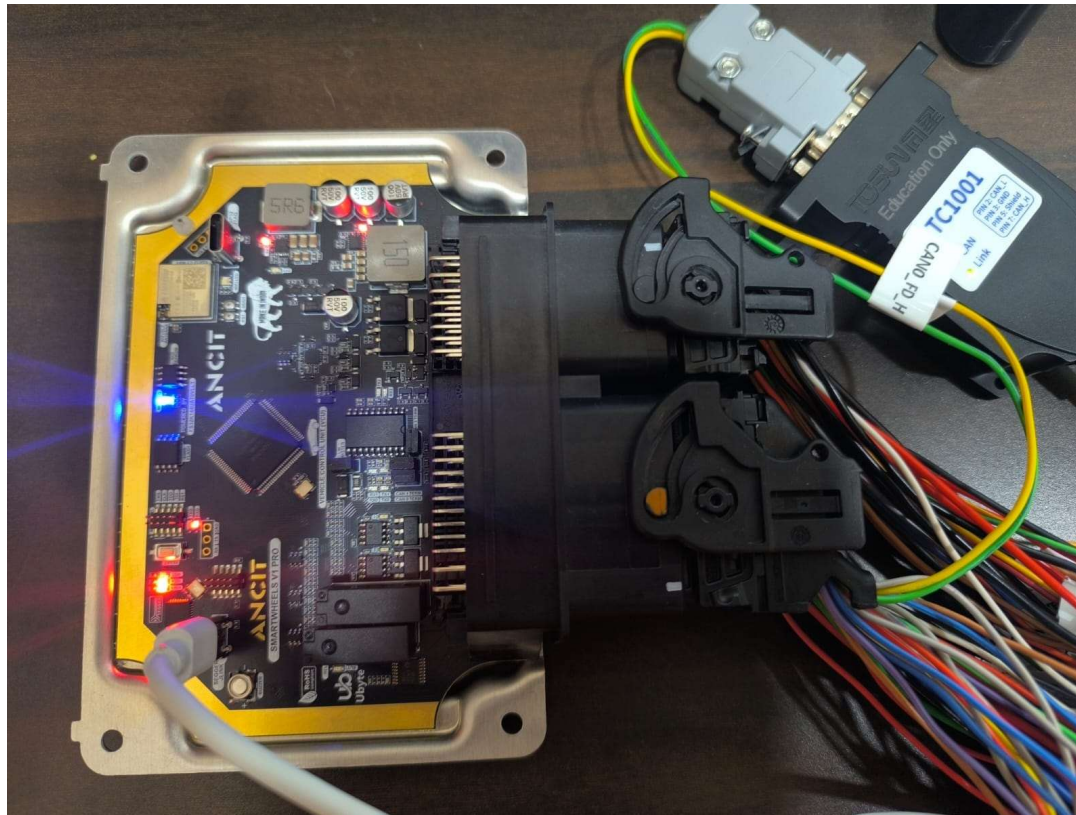


Results

- ❑ CAN byte D2 as 00 (Blue LED)
- ❑ Set the D1 byte of CAN message as FF
- ❑ And D2 byte of CAN message as 00



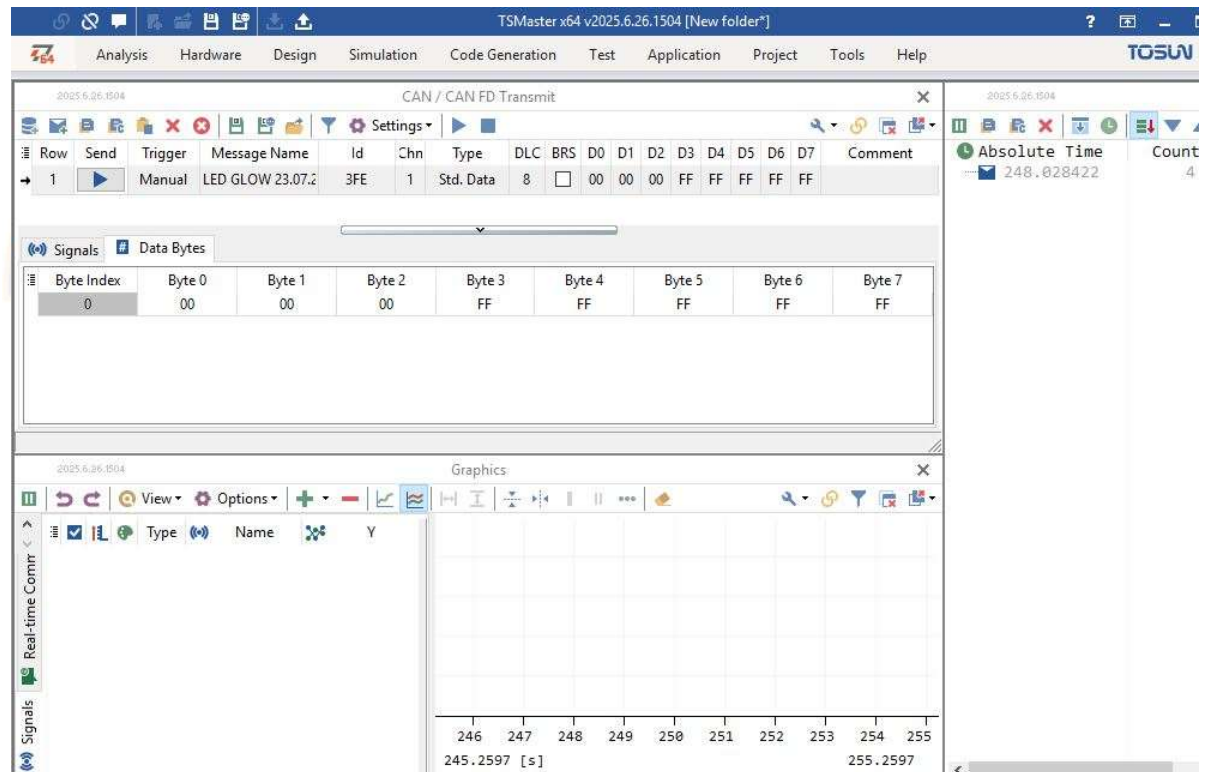
Results



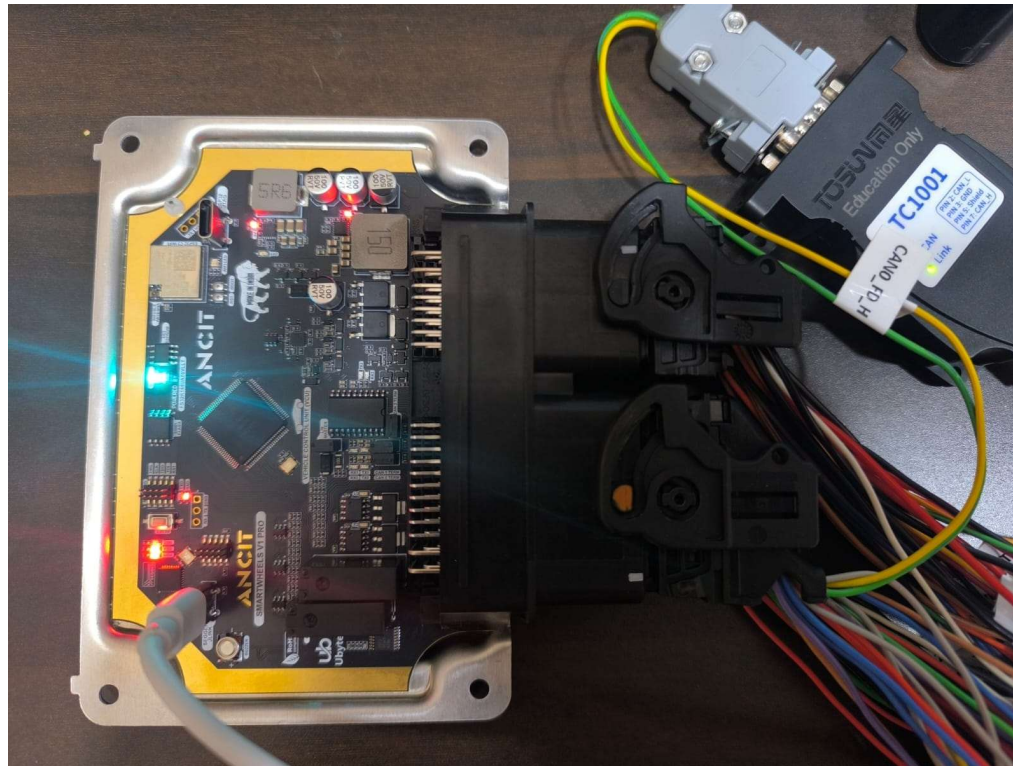
Results

❑ Set the CAN bytes D0, D1 and D2 as 00

❑ (Almost White)



Results



Thank You

